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(54) **MULTILAYER DISSOLVABLE SOLID
ARTICLE CONTAINING COATING
COMPOSITION AND PROCESS FOR
MAKING THE SAME**

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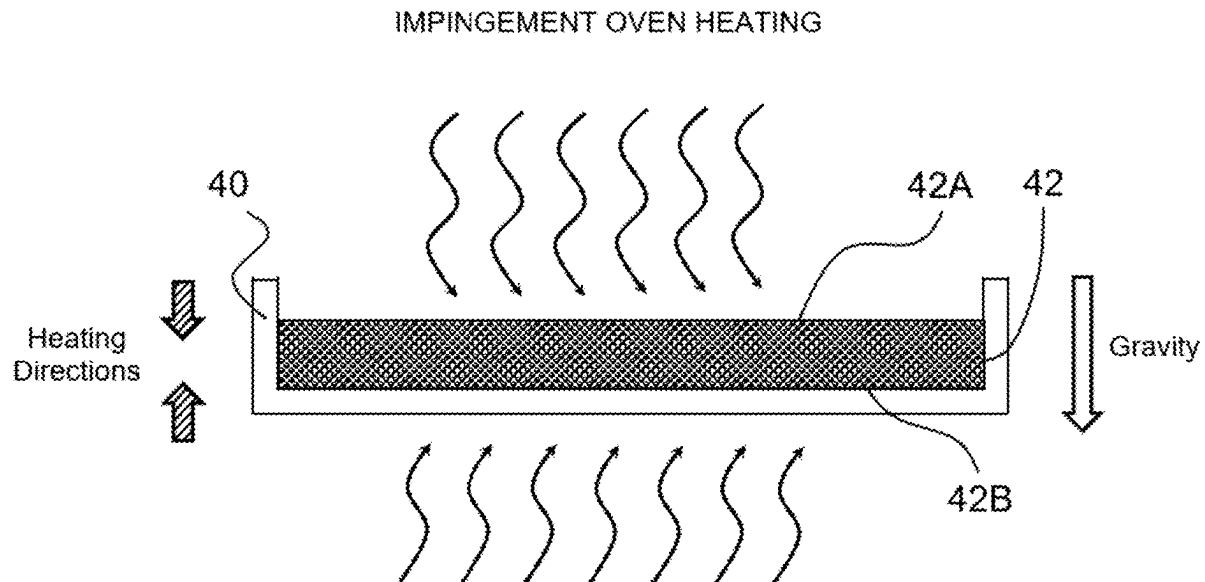
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(57)

ABSTRACT

This present invention provides dissolvable solid articles comprising multiple layers of flexible, dissolvable, porous sheets, in which a coating composition is present on at least one internal surface of at least one sheet in said solid articles. The present invention also provides a process for making such solid articles.



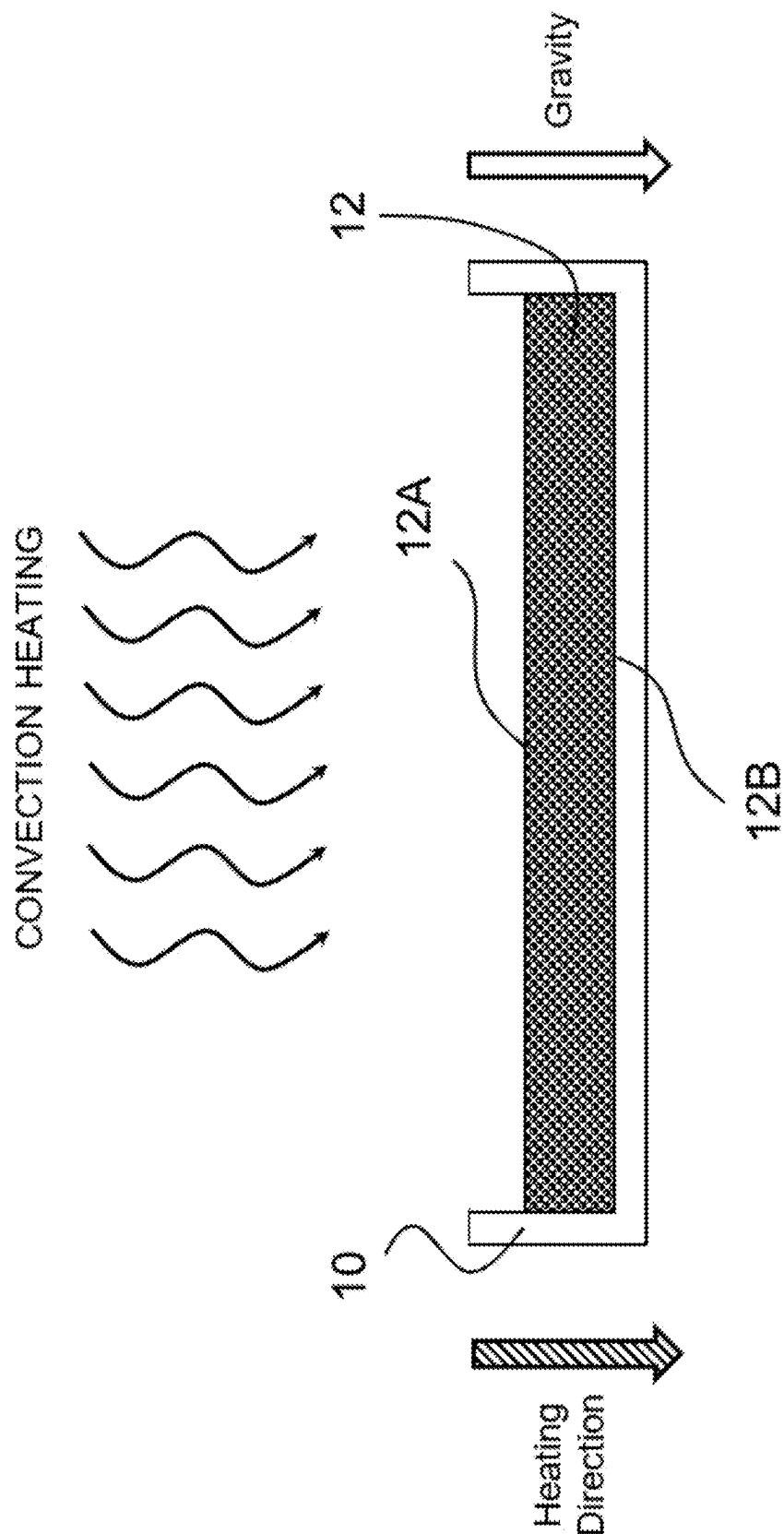


FIG. 1

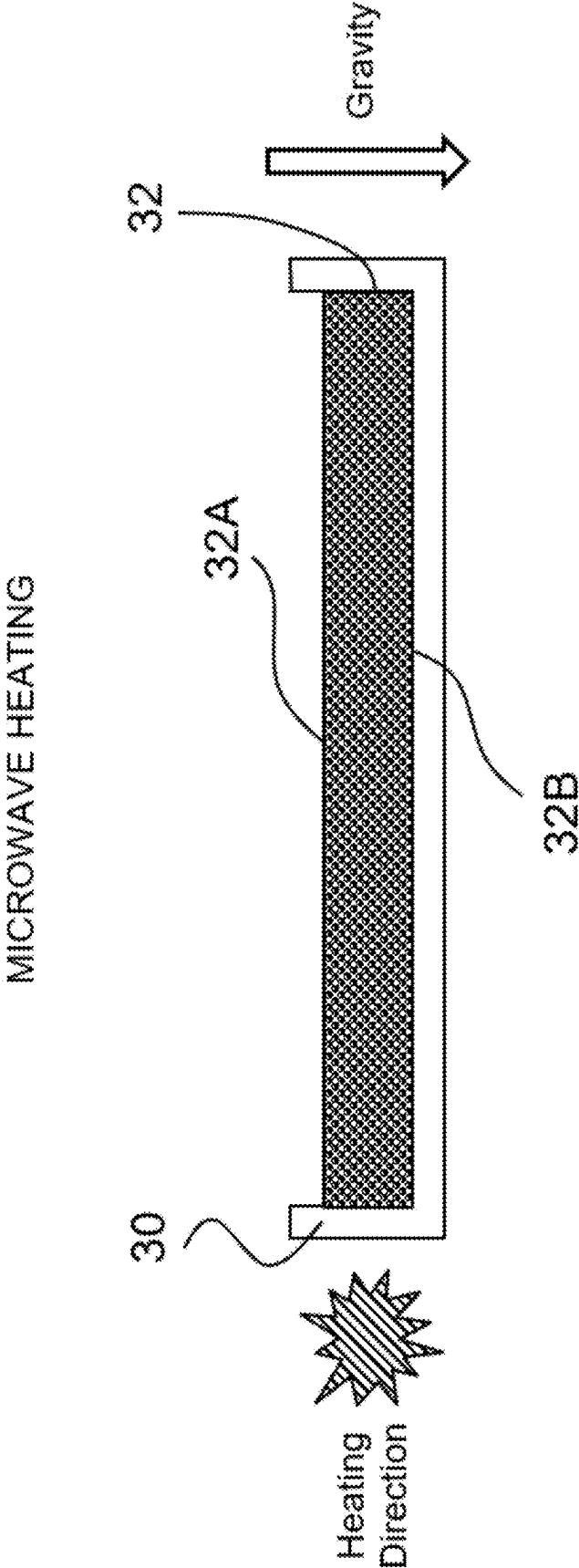


FIG. 2

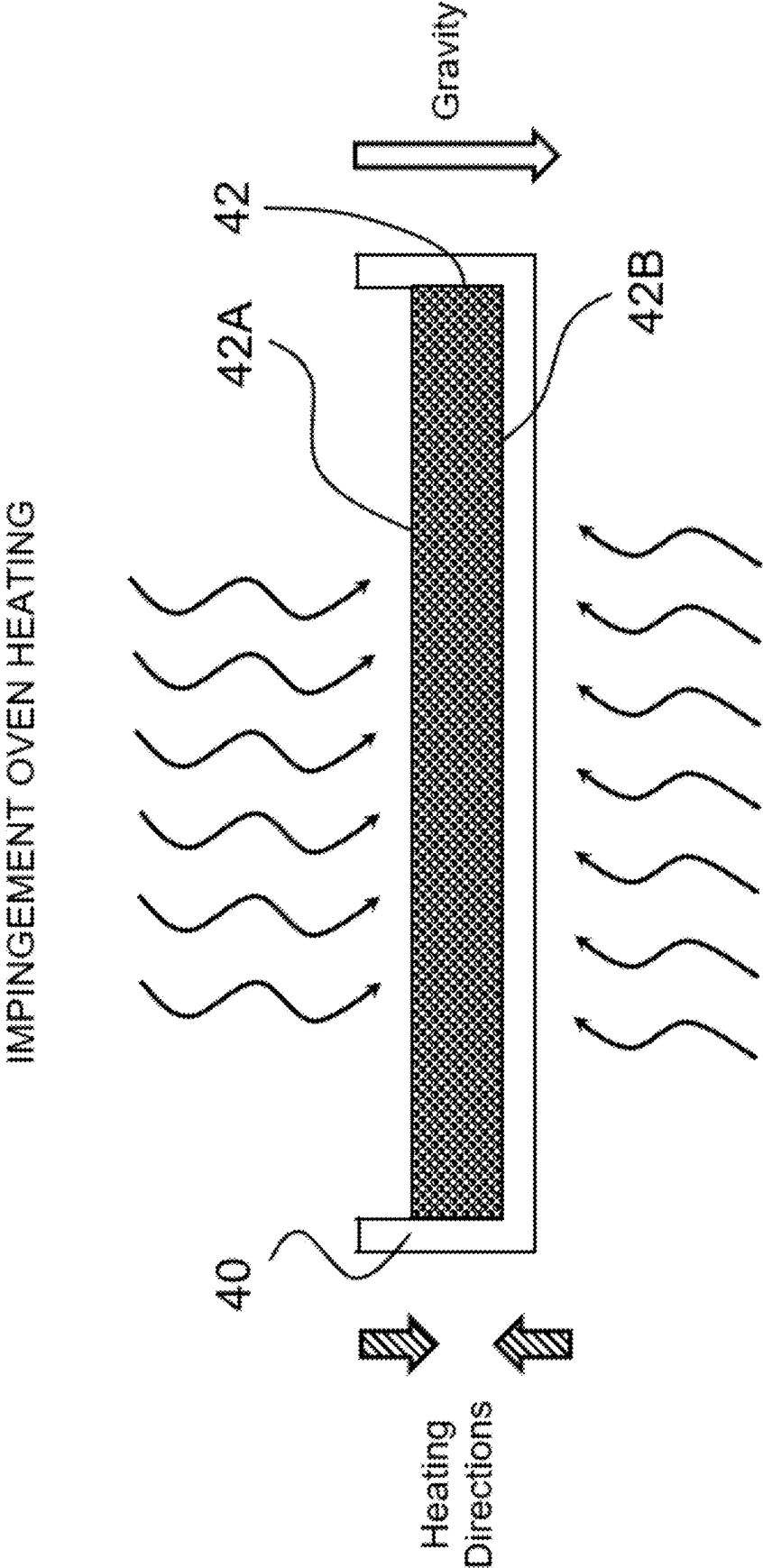


FIG. 3

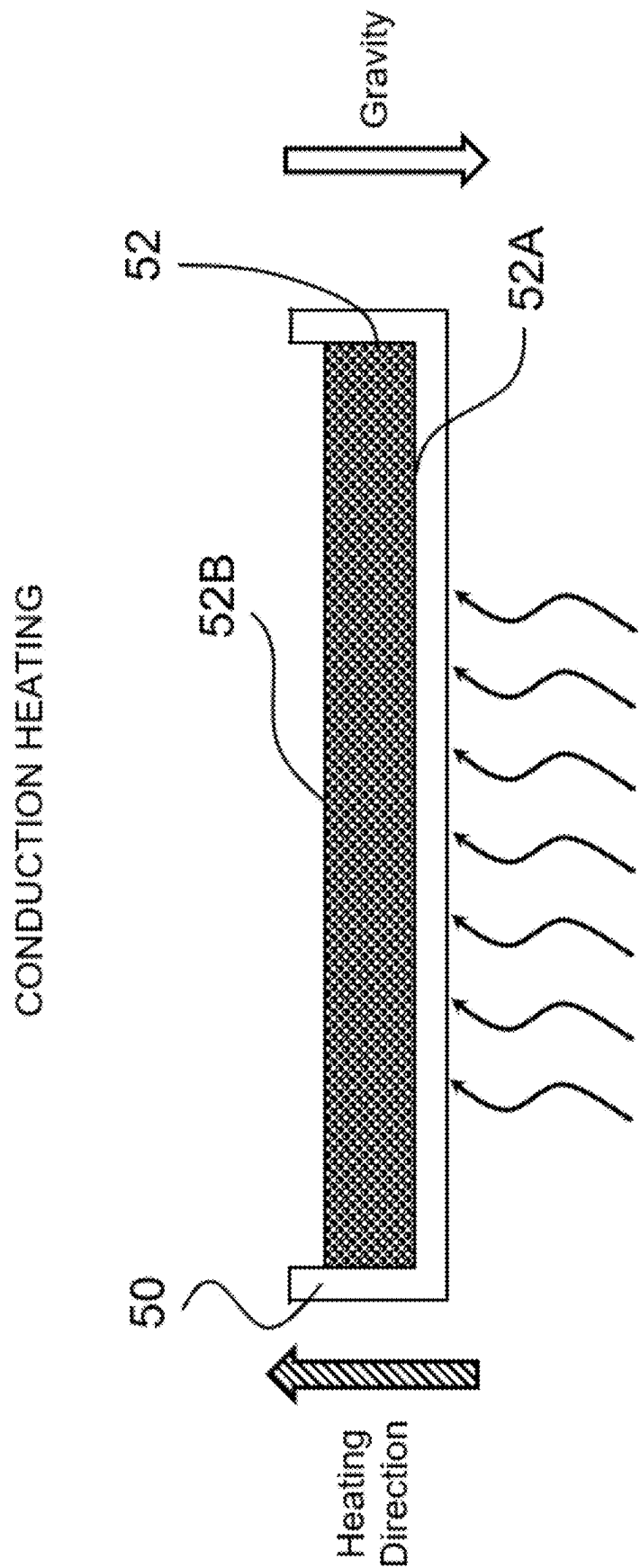


FIG. 4

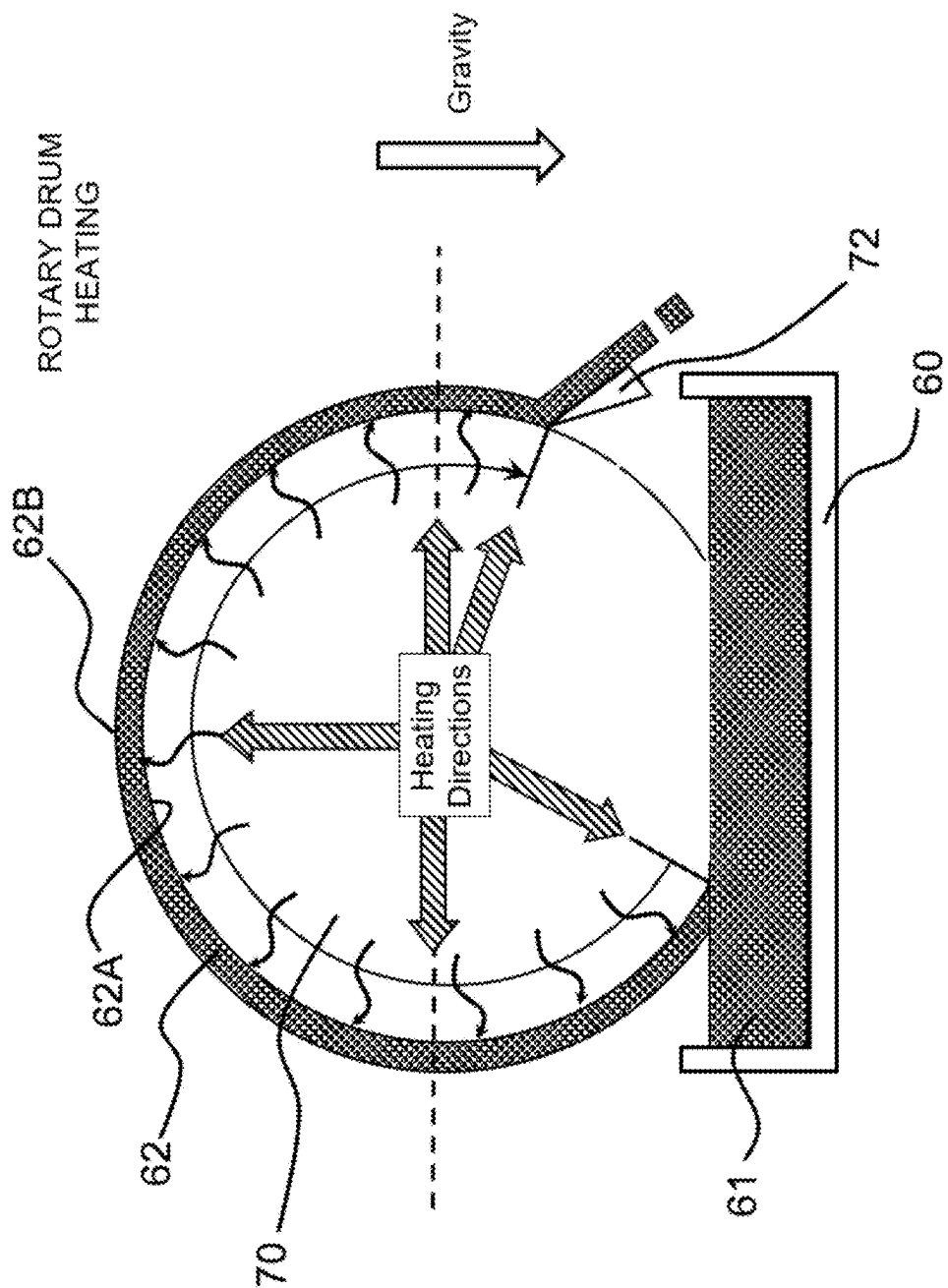


FIG. 5

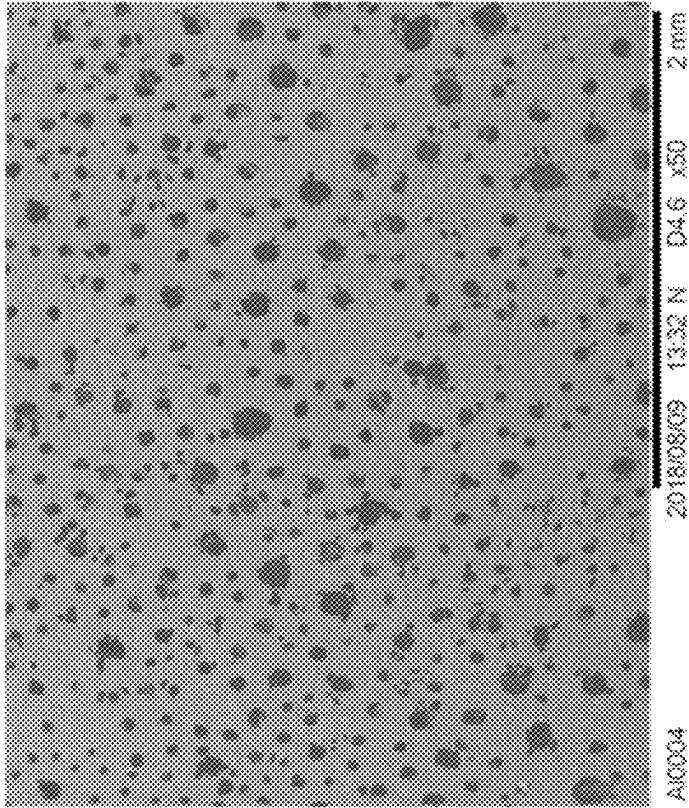


FIG. 6B
(Laundry Impingement Oven)

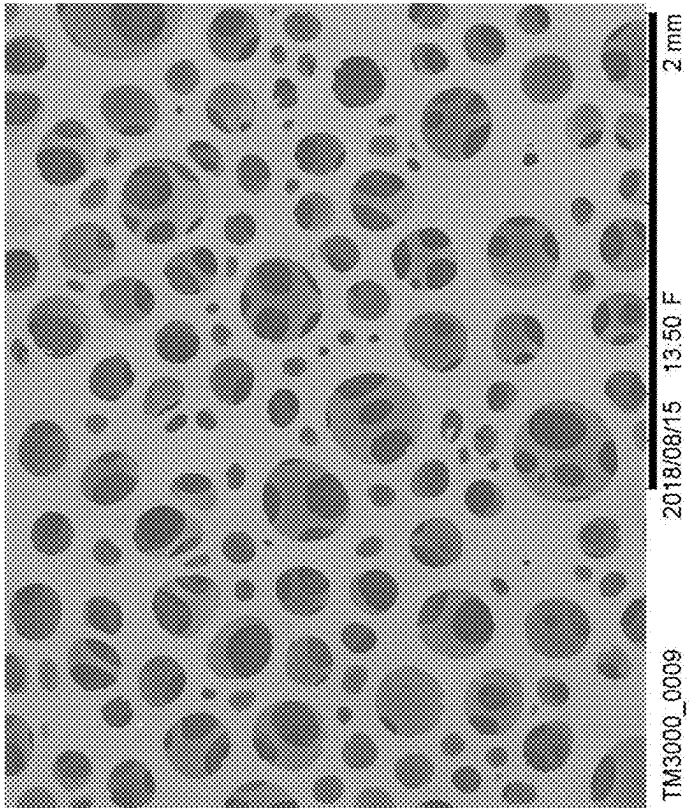


FIG. 6A
(Laundry Drum)

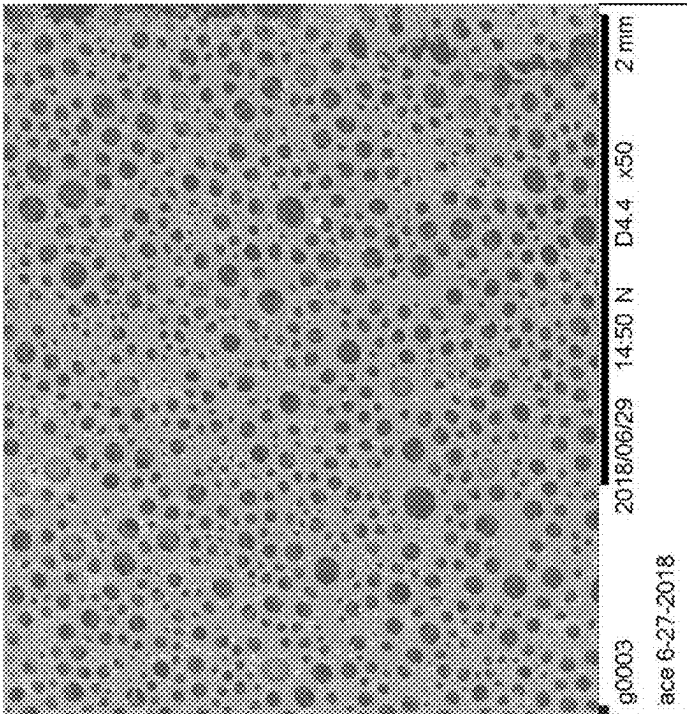


FIG. 7B
(Shampoo Impingement Oven)

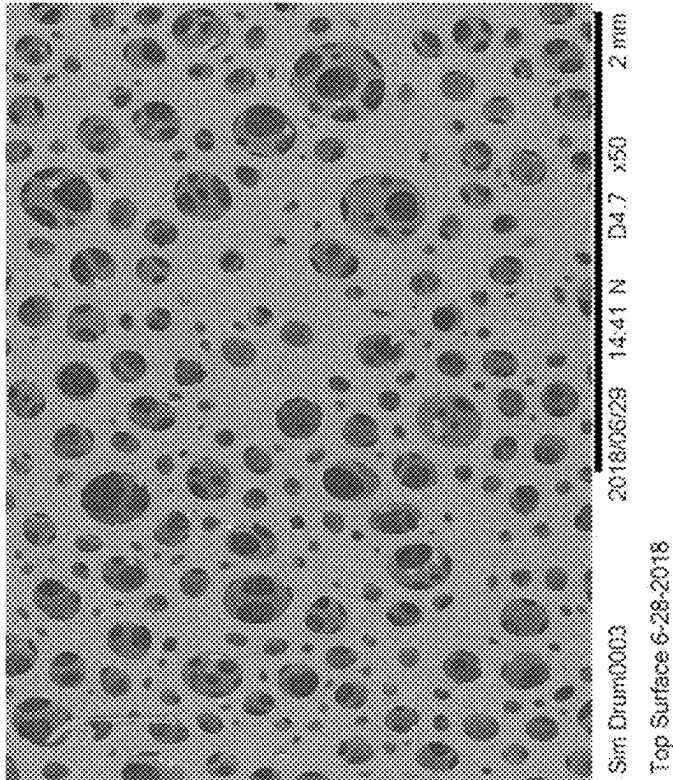


FIG. 7A
(Shampoo Hotplate)

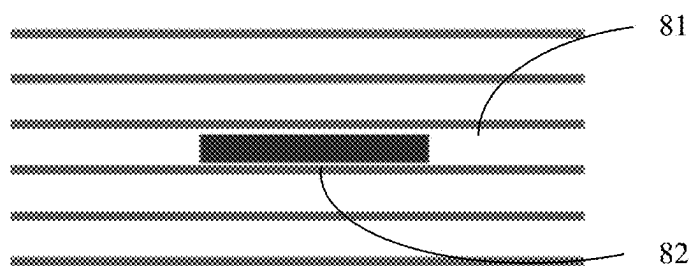


Fig. 8A

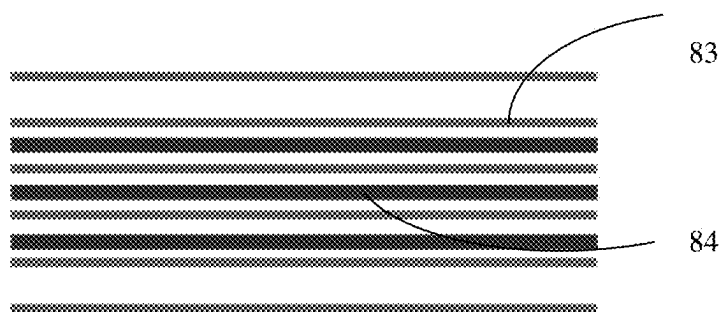


Fig. 8B

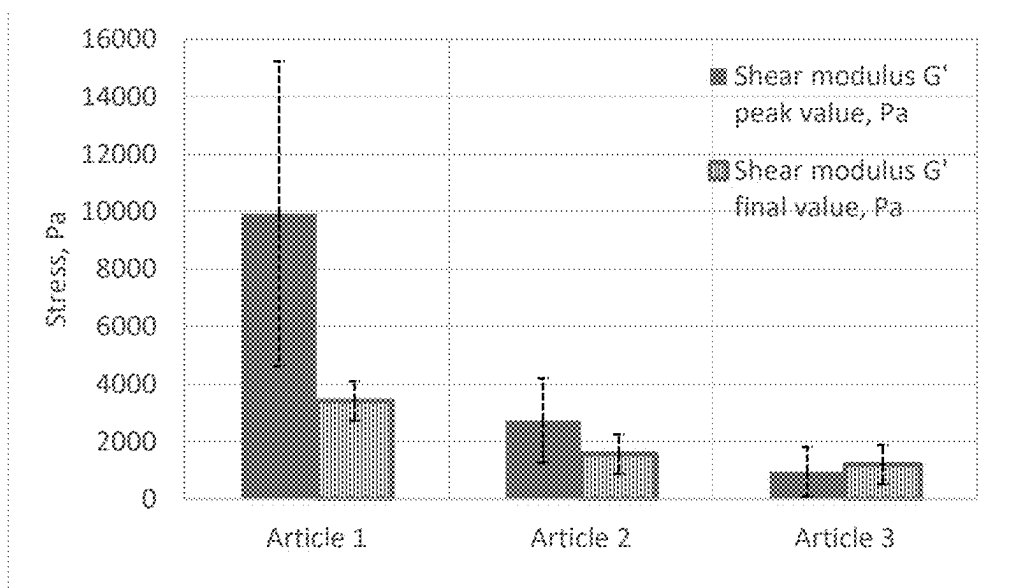


Fig. 9A

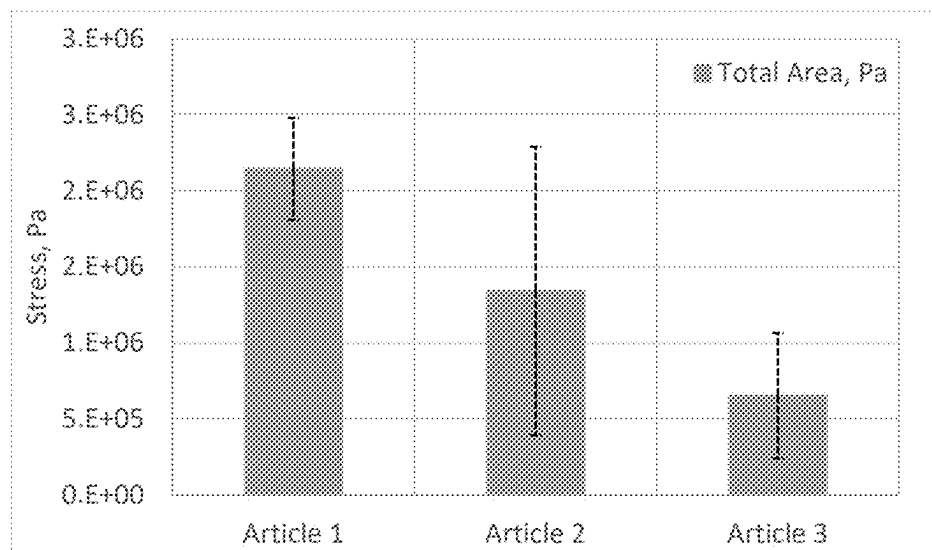


Fig. 9B

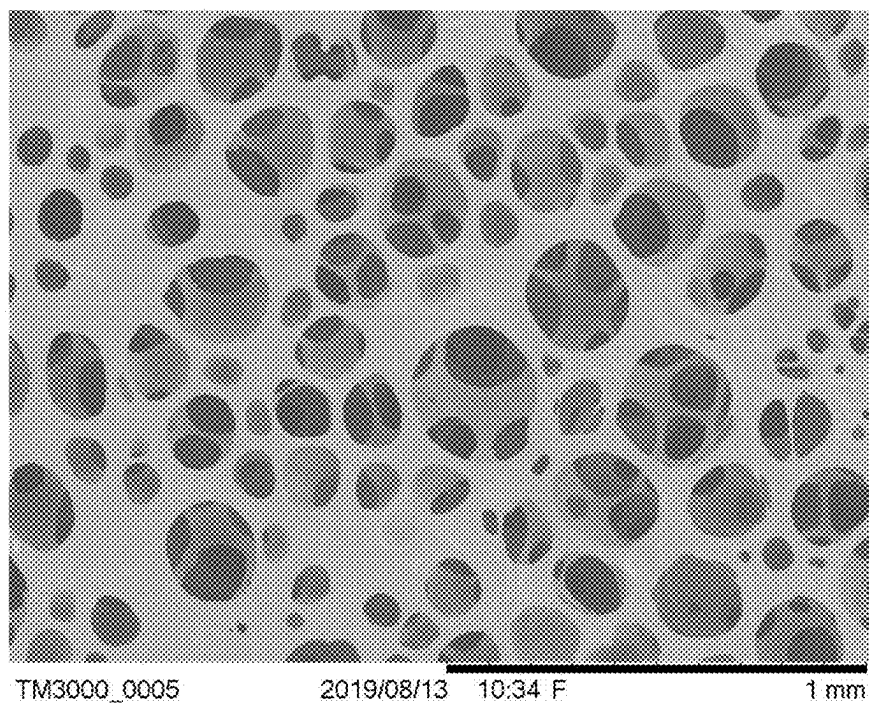


FIG. 10A

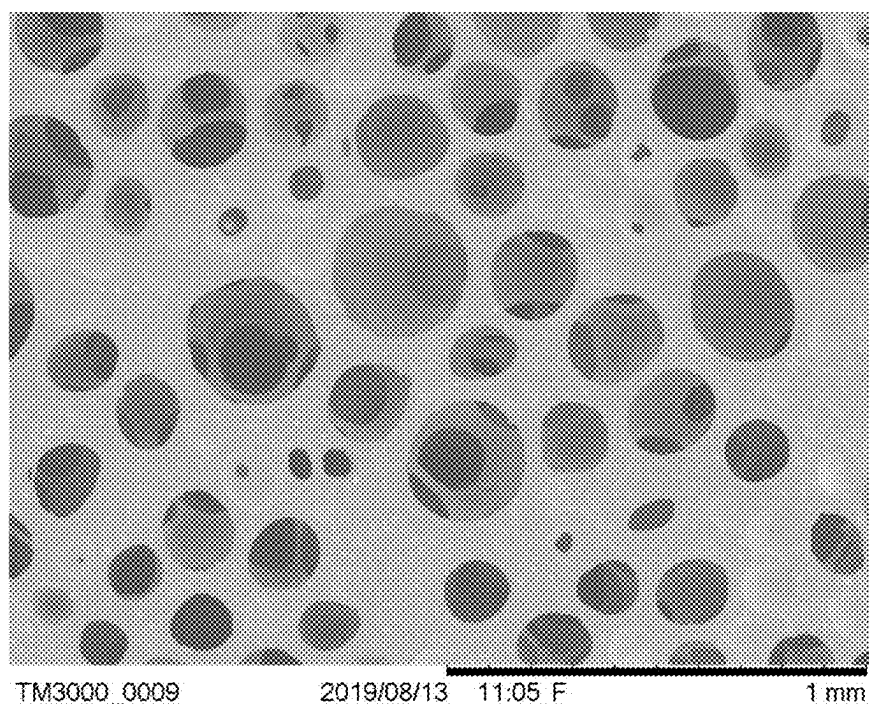


FIG. 10B

**MULTILAYER DISSOLVABLE SOLID
ARTICLE CONTAINING COATING
COMPOSITION AND PROCESS FOR
MAKING THE SAME**

FIELD OF THE INVENTION

[0001] The present invention relates to multilayer dissolvable solid articles containing a coating composition and a process for making the same.

BACKGROUND OF THE INVENTION

[0002] Flexible dissolvable sheets comprising surfactant (s) and/or other active ingredients in a water-soluble polymeric carrier or matrix are well known. Such sheets are particularly useful for delivering surfactants and/or other active ingredients upon dissolution in water. In comparison with traditional granular or liquid forms in the same product category, such sheets have better structural integrity, are more concentrated and easier to store, ship/transport, carry, and handle. In comparison with the solid tablet form in the same product category, such sheets are more flexible and less brittle, with better sensory appeal to the consumers.

[0003] In order to deliver a sufficient amount of surfactant (s) and/or other active ingredients to achieve the desired product function, it is desirable to use multiple layers of such flexible and dissolvable sheets, and it is further desirable to assemble such multiple layers into a unitary dissolvable solid article, which can then be sold as a unitary finished product. However, various challenges may be encountered when trying to assemble multiple layers of these flexible and dissolvable sheets into a unitary article, including significantly slower dissolution rate in water, in comparison with a single layer structure. In some instances, multilayered sheets may encounter an issue of gelling. Particularly, gelling occurs when the multilayered sheets are contacted with water due to the dissolution of water-soluble polymer (e.g., PVA) and surfactants in the solid articles. The presence of gelling might prevent the water to penetrate into the multilayered sheets, resulting in the reduced dissolution rate. There is also a risk that such multilayer structures may not completely dissolve under certain stringent washing conditions (e.g., cold water or extremely hard water, or low water washing conditions), and may leave undissolved residues, which can become a big consumer "pain point".

[0004] To improve dissolution, some studies have developed porous sheets with open-celled foam (OCF) structures characterized by a Percent Open Cell Content of from about 80% to 100%. Although such OCF structures significantly improve the dissolution rate of the resulting porous sheets, it is desirable for consumers to get even further improved dissolution profile including less gelling and/or less chance to leave undissolved residues.

[0005] Therefore, there is a continuing need for a multilayer structure with improved dissolution rate.

SUMMARY OF THE INVENTION

[0006] The present invention employs a coating composition applied on one or both contacting surfaces of adjacent layers of the multilayer flexible, dissolvable, porous sheets to further improve the dissolution profile of the multilayer structures. Prior to the present invention, it was believed that applying an additional component (for example, a coating composition) between layers of the multilayer flexible,

dissolvable, porous sheets likely might have a negative impact on the flow of water through the porous sheets (e.g., to block the porous structure) and thereby might adversely affect the overall dissolution profile of the sheets. Surprisingly, inventors of the present invention have unexpectedly discovered that multilayer dissolvable solid articles containing a coating composition provides a significantly improved dissolution profile.

[0007] The present invention is related, in one aspect, to a process for preparing a dissolvable solid article comprising the steps of: 1) providing two or more flexible, porous, dissolvable sheets and a coating composition, wherein each of the two or more sheets comprises a water-soluble polymer and a first surfactant and is characterized by a Percent Open Cell Content of from 80% to 100% and an Overall Average Pore Size of from 100 μm to 2000 μm , and wherein the coating composition comprises a second surfactant; 2) applying the coating composition on at least one surface of at least one sheet from the two or more sheets; and 3) arranging the two or more sheets into a stack to form the dissolvable solid article so that the coating composition is not on any of the outer surfaces of the stack.

[0008] In another aspect, the present invention is related to a dissolvable solid article comprising two or more flexible, porous, dissolvable sheets, wherein each of the two or more sheets comprises a water-soluble polymer and a first surfactant and is characterized by a Percent Open Cell Content of from 80% to 100% and an Overall Average Pore Size of from 100 μm to 2000 μm ; and wherein a coating composition comprising a second surfactant is present on at least one surface of at least one of the two or more sheets, provided that the coating composition is not on any of the outer surfaces of the dissolvable solid article. In a further aspect, the present invention is related to a dissolvable solid article comprising two or more flexible, porous, dissolvable sheets, wherein each of said two or more sheets comprises a water-soluble polymer and a first surfactant and is characterized by a Percent Open Cell Content of from 80% to 100%, an Overall Average Pore Size of from 100 μm to 2000 μm and a density of from 0.05 grams/ cm^3 to 0.17 grams/ cm^3 ; wherein a coating composition comprising a second surfactant is present on at least one surface of at least one of said two or more sheets, provided that said coating composition is not on any of the outer surfaces of the dissolvable solid article.

[0009] In a further aspect, the present invention is related to a dissolvable solid article comprising two or more flexible, porous, dissolvable sheets, wherein each of said two or more sheets comprises a water-soluble polymer and a first surfactant and is characterized by a Percent Open Cell Content of from 80% to 100% and an Overall Average Pore Size of from 100 μm to 2000 μm ; wherein a coating composition comprising a second surfactant and a solvent is present on at least one surface of at least one of said two or more sheets, provided that said coating composition is not on any of the outer surfaces of the dissolvable solid article; and wherein said solvent is selected from the group consisting of glycerol, propylene glycol, 1,3-propanediol, diethylene glycol, dipropylene glycol, ethanolamine, ethanol, water and any combinations thereof.

[0010] Preferably, the coating composition may be a liquid having a viscosity of from about 1 cps to about 25,000 cps, preferably from about 2 cps to about 10,000 cps, more preferably from about 3 cps to about 5,000 cps, most

preferably from about 1,000 cps to about 5,000 cps, as measured at about 20° C. and 1 s⁻¹. A preferred viscosity of the coating composition may provide an even better balance between the dissolution profile and the leakage.

[0011] It is an advantage of the dissolvable solid article according to the present disclosure that the dissolvable solid article containing a coating composition applied therein shows a significantly improved dissolution profile compared to the dissolvable solid article without the coating composition.

[0012] It is an advantage of the dissolvable solid article according to the present disclosure that it may function as a carrier for actives contained in the coating composition. More advantageously, it may achieve that two or more incompatible ingredients are respectively present in the sheet and the coating composition. The dissolvable solid article according to the present disclosure may have much more flexibility compared to the dissolvable solid article without the coating composition.

[0013] It is an advantage of the dissolvable solid article according to the present disclosure that the coating composition may allow more compact products with the same amount of surfactants because the coating composition has a relatively high density.

[0014] These and other aspects of the present invention will become more apparent upon reading the following detailed description of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 shows a convection-based heating/drying arrangement for making a flexible, porous, dissolvable solid sheet article in a batch process.

[0016] FIG. 2 shows a microwave-based heating/drying arrangement for making a flexible, porous, dissolvable solid sheet article in a batch process.

[0017] FIG. 3 shows an impingement oven-based heating/drying arrangement for making a flexible, porous dissolvable solid sheet article in a continuous process.

[0018] FIG. 4 shows a bottom conduction-based heating/drying arrangement for making an flexible, porous, dissolvable sheet in a batch process, according to one embodiment of the present invention.

[0019] FIG. 5 shows a rotary drum-based heating/drying arrangement for making another flexible, porous, dissolvable sheet in a continuous process, according to another embodiment of the present invention.

[0020] FIG. 6A shows a Scanning Electron Microscopic (SEM) image of the top surface of a flexible, porous, dissolvable sheet containing fabric care actives, which is made by a process employing a rotary drum-based heating/drying arrangement. FIG. 6B shows a SEM image of the top surface of an alternative flexible, porous, dissolvable sheet containing the same fabric care actives as the sheet shown in FIG. 6A, but which is made by a process employing an impingement oven-based heating/drying arrangement.

[0021] FIG. 7A shows a SEM image of the top surface of an flexible, porous, dissolvable sheet containing hair care actives, which is made by a process employing a bottom conduction-based heating/drying arrangement. FIG. 7B shows a SEM image of the top surface of an alternative flexible, porous, dissolvable sheet containing the same hair care actives as the sheet shown in FIG. 7A, but which is made by a process employing an impingement oven-based heating/drying arrangement.

[0022] FIG. 8A shows an illustrative diagram of an embodiment of the dissolvable solid article having multiple flexible, porous sheets according to the present disclosure, in which the coating composition is applied in a central region on the contacting surfaces of the middle two adjacent sheets. FIG. 8B shows an illustrative diagram of another embodiment of the dissolvable solid article having multiple flexible, porous sheets according to the present disclosure, in which the coating composition is applied throughout the contacting surfaces of any two adjacent sheets each of which is not an outermost sheet.

[0023] FIG. 9 shows results of the gelling test for solid articles containing a coating composition and solid articles without a coating composition. FIG. 9A shows shear modulus G' peak value and shear modulus G' final value; and FIG. 9B shows the total area.

[0024] FIG. 10 shows a Scanning Electron Microscopic (SEM) image of the solid sheet only (FIG. 10A) and the solid article containing a coating composition (FIG. 10B).

DETAILED DESCRIPTION OF THE INVENTION

I. Definitions

[0025] The term “flexible” as used herein refers to the ability of an article to withstand stress without breakage or significant fracture when it is bent at 90° along a center line perpendicular to its longitudinal direction. Preferably, such article can undergo significant elastic deformation and is characterized by a Young's Modulus of no more than 5 GPa, preferably no more than 1 GPa, more preferably no more than 0.5 GPa, most preferably no more than 0.2 GPa.

[0026] The term “dissolvable” as used herein refers to the ability of an article to completely or substantially dissolve in a sufficient amount of deionized water at 20° C. and under the atmospheric pressure within eight (8) hours without any stirring, leaving less than 5 wt % undissolved residues.

[0027] The term “solid” as used herein refers to the ability of an article to substantially retain its shape (i.e., without any visible change in its shape) at 20° C. and under the atmospheric pressure, when it is not confined and when no external force is applied thereto.

[0028] The term “sheet” as used herein refers to a non-fibrous structure having a three-dimensional shape, i.e., with a thickness, a length, and a width, while the length-to-thickness aspect ratio and the width-to-thickness aspect ratio are both at least about 5:1, and the length-to-width ratio is at least about 1:1. Preferably, the length-to-thickness aspect ratio and the width-to-thickness aspect ratio are both at least about 10:1, more preferably at least about 15:1, most preferably at least about 20:1; and the length-to-width aspect ratio is preferably at least about 1.2:1, more preferably at least about 1.5:1, most preferably at least about 1.618:1.

[0029] The term “contacting surfaces” of adjacent sheets as used herein refers to two surfaces that are contacting with each other when the adjacent sheets are arranged in a stack, in which the two surfaces are respectively from the two adjacent sheets. For example, the contacting surfaces may be a lower surface of an upper sheet and an upper surface of a lower sheet if the two adjacent sheets are vertically arranged as a stack.

[0030] As used herein, the term “bottom surface” refers to a surface of the flexible, porous, dissolvable solid sheet of the present invention that is immediately contacting a sup-

porting surface upon which the sheet of aerated wet pre-mixture is placed during the drying step, while the term “top surface” refers to a surface of the sheet that is opposite to the bottom surface. Further, such solid sheet can be divided into three (3) regions along its thickness, including a top region that is adjacent to its top surface, a bottom region that is adjacent to its bottom surface, and a middle region that is located between the top and bottom regions. The top, middle, and bottom regions are of equal thickness, i.e., each having a thickness that is about $\frac{1}{3}$ of the total thickness of the sheet.

[0031] As used herein, the term “outermost sheet” refers to a sheet that is adjacent to only one sheet in the multilayer dissolvable solid article of the present invention.

[0032] The term “open celled foam” or “open cell pore structure” as used herein refers to a solid, interconnected, polymer-containing matrix that defines a network of spaces or cells that contain a gas, typically a gas (such as air), without collapse of the foam structure during the drying process, thereby maintaining the physical strength and cohesiveness of the solid. The interconnectivity of the structure may be described by a Percent Open Cell Content, which is measured by Test 3 disclosed hereinafter.

[0033] The term “water-soluble” as used herein refers to the ability of a sample material to completely dissolve in or disperse into water leaving no visible solids or forming no visibly separate phase, when at least about 25 grams, preferably at least about 50 grams, more preferably at least about 100 grams, most preferably at least about 200 grams, of such material is placed in one liter (1 L) of deionized water at 20° C. and under the atmospheric pressure with sufficient stirring.

[0034] The term “aerate”, “aerating” or “aeration” as used herein refers to a process of introducing a gas into a liquid or pasty composition by mechanical and/or chemical means.

[0035] The term “heating direction” as used herein refers to the direction along which a heat source applies thermal energy to an article, which results in a temperature gradient in such article that decreases from one side of such article to the other side. For example, if a heat source located at one side of the article applies thermal energy to the article to generate a temperature gradient that decreases from the one side to an opposing side, the heating direction is then deemed as extending from the one side to the opposing side. If both sides of such article, or different sections of such article, are heated simultaneously with no observable temperature gradient across such article, then the heating is carried out in a non-directional manner, and there is no heating direction.

[0036] The term “substantially opposite to” or “substantially offset from” as used herein refers to two directions or two lines having an offset angle of 90° or more therebetween.

[0037] The term “substantially aligned” or “substantial alignment” as used herein refers to two directions or two lines having an offset angle of less than 90° therebetween.

[0038] The term “primary heat source” as used herein refers to a heat source that provides more than 50%, preferably more than 60%, more preferably more than 70%, most preferably more than 80%, of the total thermal energy absorbed by an object (e.g., the sheet of aerated wet pre-mixture according to the present invention).

[0039] The term “controlled surface temperature” as used herein refers to a surface temperature that is relatively

consistent, i.e., with less than $\pm 20\%$ fluctuations, preferably less than $\pm 10\%$ fluctuations, more preferably less than $\pm 5\%$ fluctuations.

[0040] The term “essentially free of” or “essentially free from” means that the indicated material is at the very minimal not deliberately added to the composition or product, or preferably not present at an analytically detectable level in such composition or product. It may include compositions or products in which the indicated material is present only as an impurity of one or more of the materials deliberately added to such compositions or products.

II. Overview of Processes for Making Solid Sheets

[0041] WO2010077627 discloses a batch process for forming porous sheets with open-celled foam (OCF) structures characterized by a Percent Open Cell Content of from about 80% to 100%, which functions to improve dissolution. Specifically, a pre-mixture of raw materials is first formed, which is vigorously aerated and then heat-dried in batches (e.g., in a convection oven or a microwave oven) to form the porous sheets with the desired OCF structures. Although such OCF structures significantly improve the dissolution rate of the resulting porous sheets, there is still a visibly denser and less porous bottom region with thicker cell walls in such sheets. Such high-density bottom region may negatively impact the flow of water through the sheets and thereby may adversely affect the overall dissolution rate of the sheets. When a plurality of such sheets is stacked together to form a multilayer structure, the “barrier” effect of multiple high-density bottom regions is especially augmented.

[0042] WO2012138820 discloses a similar process as that of WO2010077627, except that continuous drying of the aerated wet pre-mixture is achieved by using, e.g., an impingement oven (instead of a convection oven or a microwave oven). The OCF sheets formed by such a continuous drying process are characterized by improved uniformity/consistency in the pore structures across different regions thereof. Unfortunately, there are still rate-limiting factors in such OCF sheets, such as a top surface with relatively smaller pore openings and a top region with relatively smaller pores (i.e., a crust-like top region), which may negatively impact the flow of water therethrough and slow down the dissolution thereof.

[0043] During the drying step in the above-described processes, the OCF structures are formed under simultaneous mechanisms of water evaporation, bubble collapse, interstitial liquid drainage from the thin film bubble facings into the plateau borders between the bubbles (which generates openings between the bubbles and forms the open cells), and solidification of the pre-mixture. Various processing conditions may influence these mechanisms, e.g., solid content in the wet pre-mixture, viscosity of the wet pre-mixture, gravity, and the drying temperature, and the need to balance such processing conditions so as to achieve controlled drainage and form the desired OCF structures.

[0044] It has been a surprising and unexpected discovery of the present invention that the direction of thermal energy employed (i.e., the heating direction) during the drying step may also have a significant impact on the resulting OCF structures, in addition to the above-mentioned processing conditions.

[0045] For example, if the thermal energy is applied in a non-directional manner (i.e., there is no clear heating direc-

tion) during the drying step, or if the heating direction is substantially aligned with the gravitational direction (i.e., with an offset angle of less than 90° in between) during most of the drying step, the resulting flexible, porous, dissolvable solid sheet tends to have a top surface with smaller pore openings and greater pore size variations in different regions along the direction across its thickness. In contrast, when the heating direction is offset from the gravitation direction (i.e., with an offset angle of 90° or more therebetween) during most of the drying step, the resulting solid sheet may have a top surface with larger pore openings and reduced pore size variations in different regions along the direction across the thickness of such sheet. Correspondingly, the latter sheets are more receptive to water flowing through and are therefore more dissolvable than the former sheets.

[0046] While not being bound by any theory, it is believed that the alignment or misalignment between the heating direction and the gravitational direction during the drying step and the duration thereof may significantly affect the interstitial liquid drainage between the bubbles, and correspondingly impacting the pore expansion and pore opening in the solidifying pre-mixture and resulting in solid sheets with very different OCF structures. Such differences are illustrated more clearly by FIGS. 1-4 hereinafter.

[0047] FIG. 1 shows a convection-based heating/drying arrangement. During the drying step, a mold 10 (which can be made of any suitable materials, such as metal, ceramic or Teflon®) is filled with an aerated wet pre-mixture, which forms a sheet 12 having a first side 12A (i.e., the top side) and an opposing second side 12B (i.e., the bottom side since it is in direct contact with a supporting surface of the mold 10). Such mold 10 is placed in a 130° C. convection oven for approximately 45-46 minutes during the drying step. The convection oven heats the sheet 12 from above, i.e., along a downward heating direction (as shown by the cross-hatched arrowhead), which forms a temperature gradient in the sheet 12 that decreases from the first side 12A to the opposing second side 12B. The downward heating direction is aligned with gravitational direction (as shown by the white arrowhead), and such an aligned position is maintained throughout the entire drying time. During drying, gravity drains the liquid pre-mixture downward toward the bottom region, while the downward heating direction dries the top region first and the bottom region last. As a result, a porous solid sheet is formed with a top surface that contains numerous pores with small openings formed by gas bubbles that have not had the chance to fully expand. Such a top surface with smaller pore openings is not optimal for water ingress into the sheet, which may limit the dissolution rate of the sheet. On the other hand, the bottom region of such sheet is dense and less porous, with larger pores that are formed by fully expanded gas bubbles, but which are very few in numbers, and the cell walls between the pores in such bottom region are thick due to the downward liquid drainage effectuated by gravity. Such a dense bottom region with fewer pores and thick cell walls is a further rate-limiting factor for the overall dissolution rate of the sheet.

[0048] FIG. 2 shows a microwave-based heating/drying arrangement. During the drying step, a mold 30 is filled with an aerated wet pre-mixture, which forms a sheet 32 having a first side 32A (the top side) and an opposing second side 32B (the bottom side). Such mold 30 is then placed in a low energy density microwave applicator (not shown), which is provided by Industrial Microwave System Inc., North Caro-

lina and operated at a power of 2.0 kW, a belt speed of 1 foot per minute and a surrounding air temperature of 54.4° C. The mold 30 is placed in such microwave application for approximately 12 minutes during the drying step. Such microwave applicator heats the sheet 32 from within, without any clear or consistent heating direction. Correspondingly, no temperature gradient is formed in the sheet 32. During drying, the entire sheet 32 is simultaneously heated, or nearly simultaneously heated, although gravity (as shown by the white arrowhead) still drains the liquid pre-mixture downward toward the bottom region. As a result, the solidified sheet so formed has more uniformly distributed and more evenly sized pores, in comparison with sheet formed by the convection-based heating/drying arrangement. However, the liquid drainage under gravity force during the microwave-based drying step may still result in a dense bottom region with thick cell walls. Further, simultaneous heating of the entire sheet 32 may still limit the pore expansion and pore opening on the top surface during the drying step, and the resulting sheet may still have a top surface with relatively smaller pore openings. Further, the microwave energy heats water within the sheet 32 and causes such water to boil, which may generate bubbles of irregular sizes and form unintended dense regions with thick cell walls.

[0049] FIG. 3 shows an impingement oven-based heating/drying arrangement. During the drying step, a mold 40 is filled with an aerated wet pre-mixture, which forms a sheet 42 having a first side 42A (the top side) and an opposing second side 42B (the bottom side). Such mold 40 is then placed in a continuous impingement oven (not shown) under conditions similar to those described in Example 1, Table 2 of WO2012138820. Such continuous impingement oven heats the sheet 42 from both top and bottom at opposing and offsetting heating directions (shown by the two cross-hatched arrowheads). Correspondingly, no clear temperature gradient is formed in the sheet 42 during drying, and the entire sheet 42 is nearly simultaneously heated from both its top and bottom surfaces. Similar to the microwave-based heating/drying arrangement described in FIG. 3, gravity (as shown by the white arrowhead) continues to drain the liquid pre-mixture downward toward the bottom region in such impingement oven-based heating/drying arrangement of FIG. 4. As a result, the solidified sheet so formed has more uniformly distributed and more evenly sized pores, in comparison with sheet formed by the convection-based heating/drying arrangement. However, the liquid drainage under gravity force during the drying step may still result in a dense bottom region with thick cell walls. Further, nearly simultaneous heating of the sheet 42 from both the may still limit the pore expansion and pore opening on the top surface during the drying step, and the resulting sheet may still have a top surface with relatively smaller pore openings.

[0050] In contrast to the above-described heating/drying arrangements (convection-based, microwave-based or impingement oven-based), the present invention provides a heating/drying arrangement for drying the aerated wet pre-mixture, in which the direction of heating is purposefully configured to counteract/reduce liquid drainage caused by the gravitational force toward the bottom region (thereby reducing the density and improving pore structures in the bottom region) and to allow more time for the air bubbles near the top surface to expand during drying (thereby forming significantly larger pore openings on the top surface

of the resulting sheet). Both features function to improve overall dissolution rate of the sheet and are therefore desirable.

[0051] FIG. 4 shows a bottom conduction-based heating/drying arrangement for making an flexible, porous, dissolvable sheet, according to one embodiment of the present invention. Specifically, a mold 50 is filled with an aerated wet pre-mixture, which forms a sheet 52 having a first side 52A (i.e., the bottom side) and an opposing second side 52B (i.e., the top side). Such mold 50 is placed on a heated surface (not shown), for example, on top of a pre-heated Peltier plate with a controlled surface temperature of about 125-130° C., for approximately 30 minutes during the drying step. Heat is conducted from the heated surface at the bottom of the mold 50 through the mold to heat the sheet 52 from below, i.e., along an upward heating direction (as shown by the cross-hatched arrowhead), which forms a temperature gradient in the sheet 52 that decreases from the first side 52A (the bottom side) to the opposing second side 52B (the top side). Such an upward heating direction is opposite to the gravitational direction (as shown by the white arrowhead), and it is maintained as so throughout the entire drying time (i.e., the heating direction is opposite to the gravitational direction for almost 100% of the drying time). During drying, the gravitational force still drains the liquid pre-mixture downward toward the bottom region. However, the upward heating direction dries the sheet from bottom up, and water vapor generated by heat at the bottom region arises upward to escape from the solidifying matrix, so the downward liquid drainage toward the bottom region is significantly limited and “counteracted”/reduced by the solidifying matrix and the uprising water vapor. Correspondingly, the bottom region of the resulting dry sheet is less dense and contains numerous pores with relatively thin cell walls. Further, because the top region is the last region that is dried during this process, the air bubbles in the top region have sufficient time to expand to form significantly larger open pores at the top surface of the resulting sheet, which are particularly effective in facilitating water ingress into the sheet. Moreover, the resulting sheet has a more evenly distributed overall pore sizes throughout different regions (e.g., top, middle, bottom) thereof.

[0052] FIG. 5 shows a rotary drum-based heating/drying arrangement for making an flexible, porous, dissolvable sheet, according to another embodiment of the present invention. Specifically, a feeding trough 60 is filled with an aerated wet pre-mixture 61. A heated rotatable cylinder 70 (also referred to as a drum dryer) is placed above the feeding trough 60. The heated drum dryer 70 has a cylindrical heated outer surface characterized by a controlled surface temperature of about 130° C., and it rotates along a clock-wise direction (as shown by the thin curved line with an arrowhead) to pick up the aerated wet pre-mixture 61 from the feeding trough 60. The aerated wet pre-mixture 61 forms a thin sheet 62 over the cylindrical heated outer surface of the drum dryer 70, which rotates and dries such sheet 62 of aerated wet pre-mixture in approximately 10-15 minutes. A leveling blade (not shown) may be placed near the slurry pick-up location to ensure a consistent thickness of the sheet 62 so formed, although it is possible to control the thickness of sheet 62 simply by modulating the viscosity of the aerated wet pre-mixture 61 and the rotating speed and surface

temperature of the drum dryer 70. Once dried, the sheet 62 can then be picked up, either manually or by a scraper 72 at the end of the drum rotation.

[0053] As shown in FIG. 5, the sheet 62 formed by the aerated wet pre-mixture 61 comprises a first side 62A (i.e., the bottom side) that directly contacts the heated outer surface of the heated drum dryer 70 and an opposing second side 62B (i.e., the top side). Correspondingly, heat from the drum dryer 70 is conducted to the sheet 62 along an outward heating direction, to heat the first side 62A (the bottom side) of the sheet 62 first and then the opposing second side 62B (the top side). Such outward heating direction forms a temperature gradient in the sheet 62 that decreases from the first side 62A (the bottom side) to the opposing second side 62B (the top side). The outward heating direction is slowly and constantly changing as the drum dryer 70 rotates, but along a very clear and predictable path (as shown by the multiple outwardly extending cross-hatched arrowheads in FIG. 4). The relative position of the outward heating direction and the gravitational direction (as shown by the white arrowhead) is also slowly and constantly changing in a similar clear and predictable manner. For less than half of the drying time (i.e., when the heating direction is below the horizontal dashed line), the outward heating direction is substantially aligned with the gravitational direction with an offset angle of less than 90° in between. During majority of the drying time (i.e., when the heating direction is flushed with or above the horizontal dashed line), the outward heating direction is opposite or substantially opposite to the gravitational direction with an offset angle of 90° or more therebetween. Depending on the initial “start” coating position of the sheet 62, the heating direction can be opposite or substantially opposite to the gravitational direction for more than 55% of the drying time (if the coating starts at the very bottom of the drum dryer 70), preferably more than 60% of the drying time (if the coating starts at a higher position of the drum dryer 70, as shown in FIG. 5). Consequently, during most of the drying step this slowly rotating and changing heating direction in the rotary drum-based heating/drying arrangement can still function to limit and “counteract”/reduce the liquid drainage in sheet 62 caused by the gravitational force, resulting in improved OCF structures in the sheet so formed. The resulting sheet as dried by the heated drum dryer 70 is also characterized by a less dense bottom region with numerous more evenly sized pores, and a top surface with relatively larger pore openings. Moreover, the resulting sheet has a more evenly distributed overall pore sizes throughout different regions (e.g., top, middle, bottom) thereof.

[0054] In addition to employing the desired heating direction (i.e., in a substantially offset relation with respect to the gravitational direction) as mentioned hereinabove, it may also be desirable and even important to carefully adjust the viscosity and/or solid content of the wet pre-mixture, the amount and speed of aeration (air feed pump speed, mixing head speed, air flow rate, density of the aerated pre-mixture and the like, which may affect bubble sizes and quantities in the aerated pre-mixture and correspondingly impact the pore size/distribution/quantity/characteristics in the solidified sheet), the drying temperature and the drying time, in order to achieve optimal OCF structure in the resulting sheet according to the present invention.

[0055] More detailed descriptions of the processes for making the flexible, porous, dissolvable sheets according to

the present invention, as well as the physical and chemical characteristics of such sheets, are provided in the ensuing sections.

III. Process of Making Solid Sheets

[0056] The present invention provides a new and improved method for making flexible, porous, dissolvable solid sheets, which comprises the steps of: (a) forming a pre-mixture containing raw materials (e.g., the water-soluble polymer, active ingredients such as surfactants, and optionally a plasticizer) dissolved or dispersed in water or a suitable solvent, which is characterized by a viscosity of from about 1,000 cps to about 25,000 cps measured at about 40° C. and 1 s⁻¹; (b) aerating the pre-mixture (e.g., by introducing a gas into the wet slurry) to form an aerated wet pre-mixture; (c) forming the aerated wet pre-mixture into a sheet having opposing first and second sides; and (d) drying the formed sheet for a drying time of from 1 minute to 60 minutes at a temperature from 70° C. to 200° C. along a heating direction that forms a temperature gradient decreasing from the first side to the second side of the formed sheet, wherein the heating direction is substantially offset from the gravitational direction for more than half of the drying time, i.e., the drying step is conducted under heating along a mostly “anti-gravity” heating direction. Such a mostly “anti-gravity” heating direction can be achieved by various means, which include but are not limited to the bottom conduction-based heating/drying arrangement and the rotary drum-based heating/drying arrangement, as illustrated hereinabove in FIGS. 4 and 5 respectively.

Step (A): Preparation of Wet Pre-Mixture

[0057] The wet pre-mixture of the present invention is generally prepared by mixing solids of interest, including the water-soluble polymer, surfactant(s) and/or other benefit agents, optional plasticizer, and other optional ingredients, with a sufficient amount of water or another solvent in a pre-mix tank. The wet pre-mixture can be formed using a mechanical mixer. Mechanical mixers useful herein, include, but aren't limited to pitched blade turbines or MAXBLEND mixer (Sumitomo Heavy Industries).

[0058] It is particularly important in the present invention to adjust viscosity of the wet pre-mixture so that it is within a predetermined range of from about 1,000 cps to about 25,000 cps when measured at 40° C. and 1 s⁻¹. Viscosity of the wet pre-mixture has a significant impact on the pore expansion and pore opening of the aerated pre-mixture during the subsequent drying step, and wet pre-mixtures with different viscosities may form flexible, porous, dissolvable solid sheets of very different foam structures. On one hand, when the wet pre-mixture is too thick/viscous (e.g., having a viscosity higher than about 25,000 cps as measured at 40° C. and 1 s⁻¹), aeration of such wet pre-mixture may become more difficult. More importantly, interstitial liquid drainage from thin film bubble facings into the plateau borders of the three-dimensional foam during the subsequent drying step may be adversely affected or significantly limited. The interstitial liquid drainage during drying is believed to be critical for enabling pore expansion and pore opening in the aerated wet pre-mixture during the subsequent drying step. As a result, the flexible, porous, dissolvable solid sheet so formed thereby may have significantly smaller pores and less interconnectivity between the pores (i.e., more “closed”

pores than open pores), which render it harder for water to ingress into and egress from such sheet. On the other hand, when the wet pre-mixture is too thin/running (e.g., having a viscosity lower than about 1,000 cps as measured at 40° C. and 1 s⁻¹), the aerated wet pre-mixture may not be sufficiently stable, i.e., the air bubbles may rupture, collapse, or coalesce too quickly in the wet pre-mixture after aeration and before drying. Consequently, the resulting solid sheet may be much less porous and more dense than desired.

[0059] In one embodiment, viscosity of the wet pre-mixture ranges from about 3,000 cps to about 24,000 cps, preferably from about 5,000 cps to about 23,000 cps, more preferably from about 10,000 cps to about 20,000 cps, as measured at 40° C. and 1 sec⁻¹. The pre-mixture viscosity values are measured using a Malvern Kinexus Lab+ rheometer with cone and plate geometry (CP1/50 SR3468 SS), a gap width of 0.054 mm, a temperature of 40° C. and a shear rate of 1.0 reciprocal seconds for a period of 360 seconds.

[0060] In a preferred but not necessary embodiment, the solids of interest are present in the wet pre-mixture at a level of from about 15% to about 70%, preferably from about 20% to about 50%, more preferably from about 25% to about 45% by total weight of the wet pre-mixture. The percent solid content is the summation of the weight percentages by weight of the total processing mixture of all solid components, semi-solid components and liquid components excluding water and any obviously volatile materials such as low boiling alcohols. On one hand, if the solid content in the wet pre-mixture is too high, viscosity of the wet pre-mixture may increase to a level that will prohibit or adversely affect interstitial liquid drainage and prevent formation of the desired predominantly open-celled porous solid structure as described herein. On the other hand, if the solid content in the wet pre-mixture is too low, viscosity of the wet pre-mixture may decrease to a level that will cause bubble rupture/collapse/coalescence and more percent (%) shrinkage of the pore structures during drying, resulting in a solid sheet that is significantly less porous and denser.

[0061] Among the solids of interest in the wet pre-mixture of the present invention, there may be present from about 1% to about 75% surfactant(s), from about 0.1% to about 25% water-soluble polymer, and optionally from about 0.1% to about 25% plasticizer, by total weight of the solids. Other actives or benefit agents can also be added into the pre-mixture.

[0062] Optionally, the wet pre-mixture is pre-heated immediately prior to and/or during the aeration process at above ambient temperature but below any temperatures that would cause degradation of the components therein. In one embodiment, the wet pre-mixture is kept at an elevated temperature ranging from about 40° C. to about 100° C., preferably from about 50° C. to about 95° C., more preferably from about 60° C. to about 90° C., most preferably from about 75° C. to about 85° C. In one embodiment, the optional continuous heating is utilized before the aeration step. Further, additional heat can be applied during the aeration process to try and maintain the wet pre-mixture at such an elevated temperature. This can be accomplished via conductive heating from one or more surfaces, injection of steam or other processing means. It is believed that the act of pre-heating the wet pre-mixture before and/or during the aeration step may provide a means for lowering the viscosity of pre-mixtures comprising higher percent solids content for improved introduction of bubbles into the mixture and

formation of the desired solid sheet. Achieving higher percent solids content is desirable since it may reduce the overall energy requirements for drying. The increase of percent solids may therefore conversely lead to a decrease in water level content and an increase in viscosity. As mentioned hereinabove, wet pre-mixtures with viscosities that are too high are undesirable for the practice of the present invention. Pre-heating may effectively counteract such viscosity increase and thus allow for the manufacture of a fast dissolving sheet even when using high solid content pre-mixtures.

Step (B): Aeration of Wet Pre-Mixture

[0063] Aeration of the wet pre-mixture is conducted in order to introduce a sufficient amount of air bubbles into the wet pre-mixture for subsequent formation of the OCF structures therein upon drying. Once sufficiently aerated, the wet pre-mixture is characterized by a density that is significantly lower than that of the non-aerated wet pre-mixture (which may contain a few inadvertently trapped air bubbles) or an insufficiently aerated wet pre-mixture (which may contain some bubbles but at a much lower volume percentage and of significantly larger bubble sizes). Preferably, the aerated wet pre-mixture has a density ranging from about 0.05 g/ml to about 0.5 g/ml, preferably from about 0.08 g/ml to about 0.4 g/ml, more preferably from about 0.1 g/ml to about 0.35 g/ml, still more preferably from about 0.15 g/ml to about 0.3 g/ml, most preferably from about 0.2 g/ml to about 0.25 g/ml.

[0064] Aeration can be accomplished by either physical or chemical means in the present invention. In one embodiment, it can be accomplished by introducing a gas into the wet pre-mixture through mechanical agitation, for example, by using any suitable mechanical processing means, including but not limited to: a rotor stator mixer, a planetary mixer, a pressurized mixer, a non-pressurized mixer, a batch mixer, a continuous mixer, a semi-continuous mixer, a high shear mixer, a low shear mixer, a submerged sparger, or any combinations thereof. In another embodiment, it may be achieved via chemical means, for example, by using chemical foaming agents to provide in-situ gas formation via chemical reaction of one or more ingredients, including formation of carbon dioxide (CO₂ gas) by an effervescent system.

[0065] In a particularly preferred embodiment, it has been discovered that the aeration of the wet pre-mixture can be cost-effectively achieved by using a continuous pressurized aerator or mixer that is conventionally utilized in the foods industry in the production of marshmallows. Continuous pressurized mixers may work to homogenize or aerate the wet pre-mixture to produce highly uniform and stable foam structures with uniform bubble sizes. The unique design of the high shear rotor/stator mixing head may lead to uniform bubble sizes in the layers of the open celled foam. Suitable continuous pressurized aerators or mixers include the Morton whisk (Morton Machine Co., Motherwell, Scotland), the Oakes continuous automatic mixer (E.T. Oakes Corporation, Hauppauge, New York), the Fedco Continuous Mixer (The Peerless Group, Sidney, Ohio), the Mondo (Haas-Mondomix B.V., Netherlands), the Aeros (Aeros Industrial Equipment Co., Ltd., Guangdong Province, China), and the Preswhip (Hosokawa Micron Group, Osaka, Japan). For example, an Aeros A20 continuous aerator can be operated at a feed pump speed setting of about 300-800 (preferably at

about 500-700) with a mixing head speed setting of about 300-800 (preferably at about 400-600) and an air flow rate of about 50-150 (preferably 60-130, more preferably 80-120) respectively. For another example, an Oakes continuous automatic mixer can be operated at a mixing head speed setting of about 10-30 rpm (preferably about 15-25 rpm, more preferably about 20 rpm) with an air flow rate of about 10-30 Litres per hour (preferably about 15-25 L/hour, more preferably about 19-20 L/hour).

[0066] In another specific embodiment, aeration of the wet pre-mixture can be achieved by using the spinning bar that is a part of the rotary drum dryer, more specifically a component of the feeding trough where the wet pre-mixture is stored before it is coated onto the heated outer surface of the drum dryer and dried. The spinning bar is typically used for stirring the wet pre-mixture to preventing phase separation or sedimentation in the feeding trough during the waiting time before it is coated onto the heated rotary drum of the drum dryer. In the present invention, it is possible to operate such spinning bar at a rotating speed ranging from about 150 to about 500 rpm, preferably from about 200 to about 400 rpm, more preferably from about 250 to about 350 rpm, to mix the wet pre-mixture at the air interface and provide sufficient mechanical agitation needed for achieving the desired aeration of the wet pre-mixture.

[0067] As mentioned hereinabove, the wet pre-mixture can be maintained at an elevated temperature during the aeration process, so as to adjust viscosity of the wet pre-mixture for optimized aeration and controlled draining during drying. For example, when aeration is achieved by using the spinning bar of the rotary drum, the aerated wet pre-mixture in the feeding trough is typically maintained at about 60° C. during initial aeration by the spinning bar (while the rotary drum is stationary), and then heated to about 70° C. when the rotary drum is heated up and starts rotating.

[0068] Bubble size of the aerated wet pre-mixture assists in achieving uniform layers in the OCF structures of the resulting solid sheet. In one embodiment, the bubble size of the aerated wet pre-mixture is from about 5 to about 100 microns; and in another embodiment, the bubble size is from about 20 microns to about 80 microns. Uniformity of the bubble sizes causes the resulting solid sheets to have consistent densities.

Step (C): Sheet-Forming

[0069] After sufficient aeration, the aerated wet pre-mixture forms one or more sheets with opposing first and second sides. The sheet-forming step can be conducted in any suitable manners, e.g., by extrusion, casting, molding, vacuum-forming, pressing, printing, coating, and the like. More specifically, the aerated wet pre-mixture can be formed into a sheet by: (i) casting it into shallow cavities or trays or specially designed sheet moulds; (ii) extruding it onto a continuous belt or screen of a dryer; (iii) coating it onto the outer surface of a rotary drum dryer. Preferably, the supporting surface upon which the sheet is formed is formed by or coated with materials that are anti-corrosion, non-interacting and/or non-sticking, such as metal (e.g., steel, chromium, and the like), TEFLON®, polycarbonate, NEO-PRENE®, HDPE, LDPE, rubber, glass and the like.

[0070] Preferably, the formed sheet of aerated wet pre-mixture has a thickness ranging from a thickness ranging from 0.5 mm to 4 mm, preferably from 0.6 mm to 3.5 mm,

more preferably from 0.7 mm to 3 mm, still more preferably from 0.8 mm to 2 mm, most preferably from 0.9 mm to 1.5 mm. Controlling the thickness of such formed sheet of aerated wet pre-mixture may be important for ensuring that the resulting solid sheet has the desired OCF structures. If the formed sheet is too thin (e.g., less than 0.5 mm in thickness), many of the air bubbles trapped in the aerated wet pre-mixture will expand during the subsequent drying step to form through-holes that extend through the entire thickness of the resulting solid sheet. Such through-holes, if too many, may significantly compromise both the overall structural integrity and aesthetic appearance of the sheet. If the formed sheet is too thick, not only it will take longer to dry, but also it will result in a solid sheet with greater pore size variations between different regions (e.g., top, middle, and bottom regions) along its thickness, because the longer the drying time, the more imbalance of forces may occur through bubble rupture/collapse/coalescence, liquid drainage, pore expansion, pore opening, water evaporation, and the like. Further, multiple layers of relatively thin sheets can be assembled into three-dimensional structures of greater thickness to deliver the desired cleaning benefits or other benefits, while still providing satisfactory pore structures for fast dissolution as well as ensuring efficient drying within a relatively short drying time.

Step (D): Drying Under Anti-Gravity Heating

[0071] A key feature of the present invention is the use of an anti-gravity heating direction during the drying step, either through the entire drying time or at least through more than half of the drying time. Without being bound by any theory, it is believed that such anti-gravity heating direction may reduce or counteract excessive interstitial liquid drainage toward the bottom region of the formed sheet during the drying step. Further, because the top surface is dried last, it allows longer time for air bubbles near the top surface of the formed sheet to expand and form pore openings on the top surface (because once the wet matrix is dried, the air bubbles can no longer expand or form surface openings). Consequently, the solid sheet formed by drying with such anti-gravity heating is characterized by improved OCF structures that enables faster dissolution as well as other surprising and unexpected benefits.

[0072] In a specific embodiment, the anti-gravity heating direction is provided by a conduction-based heating/drying arrangement, either the same or similar to that illustrated by FIG. 4. For example, the aerated wet pre-mixture can be casted into a mold to form a sheet with two opposing sides. The mold can then be placed on a hot plate or a heated moving belt or any other suitable heating device with a planar heated surface characterized by a controlled surface temperature of from about 80° C. to about 170° C., preferably from about 90° C. to about 150° C., more preferably from about 100° C. to about 140° C. Thermal energy is transferred from the planar heated surface to the bottom surface of the sheet of aerated wet pre-mixture via conduction, so that solidification of the sheet starts with the bottom region and gradually moves upward to reach the top region last. In order to ensure that the heating direction is primarily anti-gravity (i.e., substantially offset from the gravitational direction) during this process, it is preferred that the heated surface is a primary heat source for the sheet during drying. If there are any other heating sources, the overall heating

direction may change accordingly. More preferably, the heated surface is the only heat source for the sheet during drying.

[0073] In another specific embodiment, the anti-gravity heating direction is provided by a rotary drum-based heating/drying arrangement, which is also referred to as drum drying or roller drying, similar to that illustrated in FIG. 5. Drum drying is one type of contact-drying methods, which is used for drying out liquids from a viscous pre-mixture of raw materials over the outer surface of a heated rotatable drum (also referred to as a roller or cylinder) at relatively low temperatures to form sheet-like articles. It is a continuous drying process particularly suitable for drying large volumes. Because the drying is conducted at relatively low temperatures via contact-heating/drying, it normally has high energy efficiency and does not adversely affect the compositional integrity of the raw materials.

[0074] The heated rotatable cylinder used in drum drying is heated internally, e.g., by steam or electricity, and it is rotated by a motorized drive installed on a base bracket at a predetermined rotational speed. The heated rotatable cylinder or drum preferably has an outer diameter ranging from about 0.5 meters to about 10 meters, preferably from about 1 meter to about 5 meters, more preferably from about 1.5 meters to about 2 meters. It may have a controlled surface temperature of from about 80° C. to about 170° C., preferably from about 90° C. to about 150° C., more preferably from about 100° C. to about 140° C. Further, such heated rotatable cylinder is rotating at a speed of from about 0.005 rpm to about 0.25 rpm, preferably from about 0.05 rpm to about 0.2 rpm, more preferably from about 0.1 rpm to about 0.18 rpm.

[0075] The heated rotatable cylinder is preferably coated with a non-stick coating on its outer surface. The non-stick coating may be overlying on the outer surface of the heated rotatable drum, or it can be fixed to a medium of the outer surface of the heated rotatable drum. The medium includes, but is not limited to, heat-resisting non-woven fabrics, heat-resisting carbon fiber, heat-resisting metal or non-metallic mesh and the like. The non-stick coating can effectively preserve structural integrity of the sheet-like article from damage during the sheet-forming process.

[0076] There is also provided a feeding mechanism on the base bracket for adding the aerated wet pre-mixture of raw materials as described hereinabove onto the heated rotatable drum, thereby forming a thin layer of the viscous pre-mixture onto the outer surface of the heated rotatable drum. Such thin layer of the pre-mixture is therefore dried by the heated rotatable drum via contact-heating/drying. The feeding mechanism includes a feeding trough installed on the base bracket, while the feeding trough has installed thereupon at least one (preferably two) feeding hopper(s), an imaging device for dynamic observation of the feeding, and an adjustment device for adjusting the position and inclination angle of the feeding hopper. By using the adjustment device to adjust the distance between the feeding hopper and the outer surface of the heated rotatable drum, the need for different thicknesses of the formed sheet-like article can be met. The adjustment device can also be used to adjust the feeding hopper to different inclination angles so as to meet the material requirements of speed and quality. The feeding trough may also include a spinning bar for stirring the wet pre-mixture therein to avoid phase separation and sedimentation before the wet pre-mixture is coated onto the outer

surface of the heated rotatable cylinder. Such spinning bar, as mentioned hereinbefore, can also be used to aerate the wet pre-mixture as needed.

[0077] There may also be a heating shield installed on the base bracket, to prevent rapid heat lost. The heating shield can also effectively save energy needed by the heated rotatable drum, thereby achieving reduced energy consumption and provide cost savings. The heating shield is a modular assembly structure, or integrated structure, and can be freely detached from the base bracket. A suction device is also installed on the heating shield for sucking the hot steam, to avoid any water condensate falling on the sheet-like article that is being formed.

[0078] There may also be an optional static scraping mechanism installed on the base bracket, for scraping or scooping up the sheet-like article already formed by the heated rotatable drum. The static scraping mechanism can be installed on the base bracket, or on one side thereof, for transporting the already formed sheet-like article downstream for further processing. The static scraping mechanism can automatically or manually move close and go away from the heated rotatable drum.

[0079] The making process of the flexible, porous, dissolvable solid sheet of the present invention is as follows. Firstly, the heated rotatable drum with the non-stick coating on the base bracket is driven by the motorized drive. Next, the adjustment device adjusts the feeding mechanism so that the distance between the feeding hopper and the outer surface of the heated rotatable drum reaches a preset value. Meanwhile, the feeding hopper adds the aerated wet pre-mixture containing all or some raw materials for making the flexible, porous, dissolvable solid sheet onto an outer surface of the heated rotatable drum, to form a thin layer of the aerated wet pre-mixture thereon with the desired thickness as described hereinabove in the preceding section. Optionally, the suction device of the heating shield sucks the hot steam generated by the heated rotatable drum. Next, the static scraping mechanism scrapes/scoops up a dried/solidified sheet, which is formed by the thin layer of aerated wet pre-mixture after it is dried by the heated rotatable drum at a relatively low temperature (e.g., 130° C.). The dried/solidified sheet can also be manually or automatically peeled off, without such static scraping mechanism and then rolled up by a roller bar.

[0080] The total drying time in the present invention depends on the formulations and solid contents in the wet pre-mixture, the drying temperature, the thermal energy influx, and the thickness of the sheet material to be dried. Preferably, the drying time is from about 1 minute to about 60 minutes, preferably from about 2 minutes to about 30 minutes, more preferably from about 2 to about 15 minutes, still more preferably from about 2 to about 10 minutes, most preferably from about 2 to about 5 minutes.

[0081] During such drying time, the heating direction is so arranged that it is substantially opposite to the gravitational direction for more than half of the drying time, preferably for more than 55% or 60% of the drying time (e.g., as in the rotary drum-based heating/drying arrangement described hereinabove), more preferably for more than 75% or even 100% of the drying time (e.g., as in the bottom conduction-based heating/drying arrangement described hereinabove). Further, the sheet of aerated wet pre-mixture can be dried under a first heating direction for a first duration and then under a second, opposite heating direction under a second

duration, while the first heating direction is substantially opposite to the gravitational direction, and while the first duration is anywhere from 51% to 99% (e.g., from 55%, 60%, 65%, 70% to 80%, 85%, 90% or 95%) of the total drying time. Such change in heating direction can be readily achieved by various other arrangements not illustrated herein, e.g., by an elongated heated belt of a serpentine shape that can rotate along a longitudinal central axis.

IV. Physical Characteristics of Solid Sheets

[0082] The flexible, porous, dissolvable solid sheet formed by the above-described processing steps is characterized by improved pore structures that allows easier water ingress into the sheet and faster dissolution of the sheet in water. Such improved pore structures are achieved mainly by adjusting various processing conditions as described hereinabove, and they are relatively independent or less influenced by the chemical formulations or the specific ingredients used for making such sheet.

[0083] In general, such solid sheet may be characterized by: (i) a Percent Open Cell Content of from about 80% to 100%, preferably from about 85% to 100%, more preferably from about 90% to 100%, as measured by the Test 3 hereinafter; and (ii) an Overall Average Pore Size of from about 100 μm to about 2000 μm , preferably from about 150 μm to about 1000 μm , more preferably from about 200 μm to about 600 μm , as measured by the Micro-CT method described in Test 2 hereinafter. The Overall Average Pore Size defines the porosity of the OCF structure of the present invention. The Percent Open Cell Content defines the interconnectivity between pores in the OCF structure of the present invention. Interconnectivity of the OCF structure may also be described by a Star Volume or a Structure Model Index (SMI) as disclosed in WO2010077627 and WO2012138820.

[0084] Such solid sheet of the present invention has opposing top and bottom surfaces, while its top surface may be characterized by a Surface Average Pore Diameter that is greater than about 100 μm , preferably greater than about 110 μm , preferably greater than about 120 μm , more preferably greater than about 130 μm , most preferably greater than about 150 μm , as measured by the SEM method described in Test 1 hereinafter. When comparing with solid sheets formed by conventional heating/drying arrangements (e.g., the convection-based, the microwave-based, or the impingement oven-based arrangements), the solid sheet formed by the improved heating/drying arrangement of the present invention has a significantly larger Surface Average Pore Diameter at its top surface (as demonstrated by FIGS. 6A-6B and 7A-7B, which are described in detail in Example 1 hereinafter), because under the specifically arranged directional heating of the present invention, the top surface of the formed sheet of aerated wet pre-mixture is the last to dry/solidify, and the air bubbles near the top surface has the longest time to expand and form larger pore openings at the top surface.

[0085] Still further, the solid sheet formed by the improved heating/drying (for example, rotary drum-based heating/drying) arrangement of the present invention is characterized by a more uniform pore size distribution between different regions along its thickness direction, in comparison with the sheets formed by other heating/drying arrangements (for example, impingement oven-based). Specifically, the solid sheet of the present invention comprises

a top region adjacent to the top surface, a bottom region adjacent to the bottom surface, and a middle region therebetween, while the top, middle, and bottom regions all have the same thickness. Each of the top, middle and bottom regions of such solid sheet is characterized by an Average Pore Size, while the ratio of Average Pore Size in the bottom region over that in the top region (i.e., bottom-to-top Average Pore Size ratio) is from about 0.6 to about 1.5, preferably from about 0.7 to about 1.4, preferably from about 0.8 to about 1.3, more preferably from about 1 to about 1.2. In comparison, a solid sheet formed by an impingement oven-based heating/drying arrangement may have a bottom-to-top Average Pore Size ratio of more than 1.5, typically about 1.7-2.2 (as demonstrated in Example 1 hereinafter). Moreover, the solid sheet of the present invention may be characterized by a bottom-to-middle Average Pore Size ratio of from about 0.5 to about 1.5, preferably from about 0.6 to about 1.3, more preferably from about 0.8 to about 1.2, most preferably from about 0.9 to about 1.1, and a middle-to-top Average Pore Size ratio of from about 1 to about 1.5, preferably from about 1 to about 1.4, more preferably from about 1 to about 1.2.

[0086] Still further, the relative standard deviation (RSTD) between Average Pore Sizes in the top, middle and bottom regions of the solid sheet of the present invention is no more than 20%, preferably no more than 15%, more preferably no more than 10%, most preferably no more than 5%. In contrast, a solid sheet formed by an impingement oven-based heating/drying arrangement may have a relative standard deviation (RSTD) between top/middle/bottom Average Pore Sizes of more than 20%, likely more than 25% or even more than 35% (as demonstrated in Example 1 hereinafter).

[0087] Preferably, the solid sheet of the present invention is further characterized by an Average Cell Wall Thickness of from about 5 μm to about 200 μm , preferably from about 10 μm to about 100 μm , more preferably from about 10 μm to about 80 μm , as measured by Test 2 hereinafter.

[0088] The solid sheet of the present invention may contain a small amount of water. Preferably, it is characterized by a final moisture content of from 0.5% to 25%, preferably from 1% to 20%, more preferably from 3% to 10%, by weight of the solid sheet, as measured by Test 4 hereinafter. An appropriate final moisture content in the resulting solid sheet may ensure the desired flexibility/deformability of the sheet, as well as providing soft/smooth sensory feel to the consumers. If the final moisture content is too low, the sheet may be too brittle or rigid. If the final moisture content is too high, the sheet may be too sticky, and its overall structural integrity may be compromised.

[0089] The solid sheet of the present invention may have a thickness ranging from about 0.6 mm to about 3.5 mm, preferably from about 0.7 mm to about 3 mm, more preferably from about 0.8 mm to about 2 mm, most preferably from about 1 mm to about 2 mm. Thickness of the solid sheet can be measured using Test 6 described hereinafter. The solid sheet after drying may be slightly thicker than the sheet of aerated wet pre-mixture, due to pore expansion that in turn leads to overall volume expansion.

[0090] The solid sheet of the present invention may further be characterized by a basis weight of from about 50 grams/ m^2 to about 500 grams/ m^2 , preferably from about 150

grams/ m^2 to about 450 grams/ m^2 , more preferably from about 250 grams/ m^2 to about 400 grams/ m^2 , as measured by Test 6 described hereinafter.

[0091] Still further, the solid sheet of the present invention may have a density ranging from about 0.05 grams/ cm^3 to about 0.5 grams/ cm^3 , preferably from about 0.06 grams/ cm^3 to about 0.4 grams/ cm^3 , more preferably from about 0.07 grams/ cm^3 to about 0.2 grams/ cm^3 , most preferably from about 0.08 grams/ cm^3 to about 0.15 grams/ cm^3 , as measured by Test 7 hereinafter. Density of the solid sheet of the present invention is lower than that of the sheet of aerated wet pre-mixture, also due to pore expansion that in turn leads to overall volume expansion.

[0092] In some embodiments, the solid sheets of the present invention may have a density of from about 0.06 grams/ cm^3 to about 0.16 grams/ cm^3 , preferably from about 0.07 grams/ cm^3 to about 0.15 grams/ cm^3 , more preferably from about 0.08 grams/ cm^3 to about 0.145 grams/ cm^3 . The solid article containing sheets with such relatively low density may achieve even more improved leakage performance.

[0093] Furthermore, the solid sheet of the present invention can be characterized by a Specific Surface Area of from about 0.03 m^2/g to about 0.25 m^2/g , preferably from about 0.04 m^2/g to about 0.22 m^2/g , more preferably from 0.05 m^2/g to 0.2 m^2/g , most preferably from 0.1 m^2/g to 0.18 m^2/g , as measured by Test 8 described hereinafter. The Specific Surface Area of the solid sheet of the present invention may be indicative of its porosity and may impact its dissolution rate, e.g., the greater the Specific Surface Area, the more porous the sheet and the faster its dissolution rate.

[0094] In a preferred embodiment, the solid sheet according to the present disclosure and/or the dissolvable solid article according to the present disclosure is characterized by:

[0095] a Percent Open Cell Content of from 85% to 100%, preferably from 90% to 100%; and/or

[0096] an Overall Average Pore Size of from 150 μm to 1000 μm , preferably from 200 μm to 600 μm ; and/or

[0097] an Average Cell Wall Thickness of from 5 μm to 200 μm , preferably from 10 μm to 100 μm , more preferably from 10 μm to 80 μm ; and/or

[0098] a final moisture content of from 0.5% to 25%, preferably from 1% to 20%, more preferably from 3% to 10%, by weight of the solid sheet article; and/or

[0099] a thickness of from 0.6 mm to 3.5 mm, preferably from 0.7 mm to 3 mm, more preferably from 0.8 mm to 2 mm, most preferably from 1 mm to 2 mm; and/or

[0100] a basis weight of from about 50 grams/ m^2 to about 500 grams/ m^2 , preferably from about 150 grams/ m^2 to about 450 grams/ m^2 , more preferably from about 250 grams/ m^2 to about 400 grams/ m^2 ; and/or

[0101] a density of from 0.05 grams/ cm^3 to 0.5 grams/ cm^3 , preferably from 0.06 grams/ cm^3 to 0.4 grams/ cm^3 , more preferably from 0.07 grams/ cm^3 to 0.2 grams/ cm^3 , most preferably from 0.08 grams/ cm^3 to 0.15 grams/ cm^3 ; and/or

[0102] a Specific Surface Area of from 0.03 m^2/g to 0.25 m^2/g , preferably from 0.04 m^2/g to 0.22 m^2/g , more preferably from 0.05 m^2/g to 0.2 m^2/g , most preferably from 0.1 m^2/g to 0.18 m^2/g .

V. Formulations of Solid Sheets

1. Water-Soluble Polymer

[0103] As mentioned hereinabove, the flexible, porous, dissolvable solid sheet of the present invention may be formed by a wet pre-mixture that comprises a water-soluble polymer and a first surfactant. Such a water-soluble polymer may function in the resulting solid sheet as a film-former, a structurant as well as a carrier for other active ingredients (e.g., surfactants, emulsifiers, builders, chelants, perfumes, colorants, and the like).

[0104] Preferably, the wet pre-mixture may comprise from about 3% to about 20% by weight of the pre-mixture of water-soluble polymer, in one embodiment from about 5% to about 15% by weight of the pre-mixture of water-soluble polymer, in one embodiment from about 7% to about 10% by weight of the pre-mixture of water-soluble polymer.

[0105] After drying, it is preferred that the water-soluble polymer is present in the flexible, porous, dissolvable solid sheet of the present invention in an amount ranging from about 5% to about 50%, preferably from about 8% to about 40%, more preferably from about 10% to about 30%, most preferably from about 11% to about 25%, by total weight of the solid sheet. In a particularly preferred embodiment of the present invention, the total amount of water-soluble polymer (s) present in the flexible, porous, dissolvable solid sheet of the present invention is no more than 25% by total weight of such sheet.

[0106] Water-soluble polymers suitable for the practice of the present invention may be selected those with weight average molecular weights ranging from about 50,000 to about 400,000 Daltons, preferably from about 60,000 to about 300,000 Daltons, more preferably from about 70,000 to about 200,000 Daltons, most preferably from about 80,000 to about 150,000 Daltons. The weight average molecular weight is computed by summing the average molecular weights of each polymer raw material multiplied by their respective relative weight percentages by weight of the total weight of polymers present within the porous solid sheet. The weight average molecular weight of the water-soluble polymer used herein may impact the viscosity of the wet pre-mixture, which may in turn influence the bubble number and size during the aeration step as well as the pore expansion/opening results during the drying step. Further, the weight average molecular weight of the water-soluble polymer may affect the overall film-forming properties of the wet pre-mixture and its compatibility/incompatibility with certain surfactants.

[0107] The water-soluble polymers of the present invention may include, but are not limited to, synthetic polymers including polyvinyl alcohols, polyvinylpyrrolidones, polyalkylene oxides, polyacrylates, caprolactams, polymethacrylates, polymethylmethacrylates, polyacrylamides, polymethylacrylamides, polydimethylacrylamides, polyethylene glycol monomethacrylates, copolymers of acrylic acid and methyl acrylate, polyurethanes, polycarboxylic acids, polyvinyl acetates, polyesters, polyamides, polyamines, polyethyleneimines, maleic(acrylate or methacrylate) copolymers, copolymers of methylvinyl ether and of maleic anhydride, copolymers of vinyl acetate and crotonic acid, copolymers of vinylpyrrolidone and of vinyl acetate, copolymers of vinylpyrrolidone and of caprolactam, vinyl pyrrolidone/vinyl acetate copolymers, copolymers of anionic, cationic and amphoteric monomers, and combinations thereof.

[0108] The water-soluble polymers of the present invention may also be selected from naturally sourced polymers including those of plant origin examples of which include karaya gum, tragacanth gum, gum Arabic, acemannan, konjac mannan, acacia gum, gum ghatti, whey protein isolate, and soy protein isolate; seed extracts including guar gum, locust bean gum, quince seed, and psyllium seed; seaweed extracts such as Carrageenan, alginates, and agar; fruit extracts (pectins); those of microbial origin including xanthan gum, gellan gum, pullulan, hyaluronic acid, chondroitin sulfate, and dextran; and those of animal origin including casein, gelatin, keratin, keratin hydrolysates, sulfonic keratins, albumin, collagen, glutelin, glucagons, gluten, zein, and shellac.

[0109] Modified natural polymers can also be used as water-soluble polymers in the present invention. Suitable modified natural polymers include, but are not limited to, cellulose derivatives such as hydroxypropylmethylcellulose, hydroxymethylcellulose, hydroxyethylcellulose, methylcellulose, hydroxypropylcellulose, ethylcellulose, carboxymethylcellulose, cellulose acetate phthalate, nitrocellulose and other cellulose ethers/esters; and guar derivatives such as hydroxypropyl guar.

[0110] The water-soluble polymer of the present invention may include starch. As used herein, the term “starch” include both naturally occurring or modified starches. Typical natural sources for starches can include cereals, tubers, roots, legumes and fruits. More specific natural sources can include corn, pea, potato, banana, barley, wheat, rice, sago, amaranth, tapioca, arrowroot, canna, sorghum, and waxy or high amylase varieties thereof. The natural starches can be modified by any modification method known in the art to form modified starches, including physically modified starches, such as sheared starches or thermally-inhibited starches; chemically modified starches, such as those which have been cross-linked, acetylated, and organically esterified, hydroxyethylated, and hydroxypropylated, phosphorylated, and inorganically esterified, cationic, anionic, non-ionic, amphoteric and zwitterionic, and succinate and substituted succinate derivatives thereof; conversion products derived from any of the starches, including fluidity or thin-boiling starches prepared by oxidation, enzyme conversion, acid hydrolysis, heat or acid dextrinization, thermal and or sheared products may also be useful herein; and pregelatinized starches which are known in the art.

[0111] Preferred water-soluble polymers of the present invention include polyvinyl alcohols, polyvinylpyrrolidones, polyalkylene oxides, starch and starch derivatives, pullulan, gelatin, hydroxypropylmethylcelluloses, methylcelluloses, and carboxymethylcelluloses. More preferred water-soluble polymers of the present invention include polyvinyl alcohols, and hydroxypropylmethylcelluloses.

[0112] Most preferred water-soluble polymers of the present invention are polyvinyl alcohols characterized by a degree of hydrolysis ranging from about 40% to about 100%, preferably from about 50% to about 95%, more preferably from about 65% to about 92%, most preferably from about 70% to about 90%. Commercially available polyvinyl alcohols include those from Celanese Corporation (Texas, USA) under the CELVOL trade name including, but not limited to, CELVOL 523, CELVOL 530, CELVOL 540, CELVOL 518, CELVOL 513, CELVOL 508, CELVOL 504; those from Kuraray Europe GmbH (Frankfurt, Germany) under the Mowiol® and POVAL™ trade names; and PVA

1788 (also referred to as PVA BP17) commercially available from various suppliers including Lubon Vynlon Co. (Nanjing, China); and combinations thereof. In a particularly preferred embodiment of the present invention, the flexible, porous, dissolvable solid sheet comprises from about 10% to about 25%, more preferably from about 15% to about 23%, by total weight of such sheet, of a polyvinyl alcohol having a weight average molecular weight ranging from 80,000 to about 150,000 Daltons and a degree of hydrolysis ranging from about 80% to about 90%.

[0113] In addition to polyvinyl alcohols as mentioned hereinabove, a single starch or a combination of starches may be used as a filler material in such an amount as to reduce the overall level of water-soluble polymers required, so long as it helps provide the solid sheet with the requisite structure and physical/chemical characteristics as described herein. However, too much starch may comprise the solubility and structural integrity of the sheet. Therefore, in preferred embodiments of the present invention, it is desired that the solid sheet comprises no more than 20%, preferably from 0% to 10%, more preferably from 0% to 5%, most preferably from 0% to 1%, by weight of the solid sheet, of starch.

2. First Surfactants

[0114] In addition to the water-soluble polymer described hereinabove, the solid sheet of the present invention comprises a first surfactant. The first surfactant may function as emulsifying agents during the aeration process to create a sufficient amount of stable bubbles for forming the desired OCF structure of the present invention. Further, the first surfactant may function as active ingredients for delivering a desired cleansing benefit.

[0115] In a preferred embodiment of the present invention, the solid sheet comprises a first surfactant selected from the group consisting of anionic surfactants, nonionic surfactants, cationic surfactants, zwitterionic surfactants, amphoteric surfactants, polymeric surfactants and any combinations thereof. Depending on the desired application of such solid sheet and the desired consumer benefit to be achieved, different surfactants can be selected. One benefit of the present invention is that the OCF structures of the solid sheet allow for incorporation of a high surfactant content while still providing fast dissolution. Consequently, highly concentrated cleansing compositions can be formulated into the solid sheets of the present invention to provide a new and superior cleansing experience to the consumers.

[0116] The first surfactant as used herein may include both surfactants from the conventional sense (i.e., those providing a consumer-noticeable lathering effect) and emulsifiers (i.e., those that do not provide any lathering performance but are intended primarily as a process aid in making a stable foam structure). Examples of emulsifiers for use as a surfactant component herein include mono- and di-glycerides, fatty alcohols, polyglycerol esters, propylene glycol esters, sorbitan esters and other emulsifiers known or otherwise commonly used to stabilize air interfaces.

[0117] The total amount of the first surfactant present in the solid sheet of the present invention may range widely from about 5% to about 80%, preferably from about 10% to about 70%, more preferably from about 30% to about 65%, by total weight of the solid sheet. Correspondingly, the wet pre-mixture may comprise from about 1% to about 40% by weight of the wet pre-mixture of surfactant(s), in one

embodiment from about 2% to about 35% by weight of the wet pre-mixture of surfactant(s), in one embodiment from about 5% to about 30% by weight of the wet pre-mixture of surfactant(s).

[0118] In a preferred embodiment of the present invention, the solid sheet of the present invention comprises from about 30% to about 90%, preferably from about 40% to about 80%, more preferably from about 50% to about 70%, of a first surfactant by total weight of the solid sheet. In such cases, the wet pre-mixture may comprise from about 10% to about 40% by weight of the wet pre-mixture of surfactant(s), in one embodiment from about 12% to about 35% by weight of the wet pre-mixture of surfactant(s), in one embodiment from about 15% to about 30% by weight of the wet pre-mixture of surfactant(s).

[0119] Non-limiting examples of anionic surfactants suitable for use herein include alkyl and alkyl ether sulfates, sulfated monoglycerides, sulfonated olefins, alkyl aryl sulfonates, primary or secondary alkane sulfonates, alkyl sulfosuccinates, acyl taurates, acyl isethionates, alkyl glyceryl ether sulfonate, sulfonated methyl esters, sulfonated fatty acids, alkyl phosphates, acyl glutamates, acyl sarcosinates, alkyl sulfoacetates, acylated peptides, alkyl ether carboxylates, acyl lactylates, anionic fluorosurfactants, sodium lauroyl glutamate, and combinations thereof.

[0120] One category of anionic surfactants particularly suitable for practice of the present invention include C₆-C₂₀ linear alkylbenzene sulphonate (LAS) surfactant. LAS surfactants are well known in the art and can be readily obtained by sulfonating commercially available linear alkylbenzenes. Exemplary C₁₀-C₂₀ linear alkylbenzene sulfonates that can be used in the present invention include alkali metal, alkaline earth metal or ammonium salts of C₁₀-C₂₀ linear alkylbenzene sulfonic acids, and preferably the sodium, potassium, magnesium and/or ammonium salts of C₁₁-C₁₈ or C₁-C₁₄ linear alkylbenzene sulfonic acids. More preferred are the sodium or potassium salts of C₁₂ and/or C₁₄ linear alkylbenzene sulfonic acids, and most preferred is the sodium salt of C₁₂ and/or C₁₄ linear alkylbenzene sulfonic acid, i.e., sodium dodecylbenzene sulfonate or sodium tetradecylbenzene sulfonate.

[0121] LAS provides superior cleaning benefit and is especially suitable for use in laundry detergent applications. It has been a surprising and unexpected discovery of the present invention that when polyvinyl alcohol having a higher weight average molecular weight (e.g., from about 50,000 to about 400,000 Daltons, preferably from about 60,000 to about 300,000 Daltons, more preferably from about 70,000 to about 200,000 Daltons, most preferably from about 80,000 to about 150,000 Daltons) is used as the film-former and carrier, LAS can be used as a major surfactant, i.e., present in an amount that is more than 50% by weight of the total surfactant content in the solid sheet, without adversely affecting the film-forming performance and stability of the overall composition. Correspondingly, in a particular embodiment of the present invention, LAS is used as the major surfactant in the solid sheet. If present, the amount of LAS in the solid sheet of the present invention may range from about 10% to about 70%, preferably from about 20% to about 65%, more preferably from about 40% to about 60%, by total weight of the solid sheet.

[0122] Another category of anionic surfactants suitable for practice of the present invention include sodium tridecyl sulfates (STS) having a weight average degree of alkoxy-

lation ranging from about 0.5 to about 5, preferably from about 0.8 to about 4, more preferably from about 1 to about 3, most preferably from about 1.5 to about 2.5. Trideceth is a 13-carbon branched alkoxyated hydrocarbon comprising, in one embodiment, an average of at least 1 methyl branch per molecule. STS used by the present invention may be include ST(EOxPOy)S, while EOx refers to repeating ethylene oxide units with a repeating number x ranging from 0 to 5, preferably from 1 to 4, more preferably from 1 to 3, and while POy refers to repeating propylene oxide units with a repeating number y ranging from 0 to 5, preferably from 0 to 4, more preferably from 0 to 2. It is understood that a material such as ST2S with a weight average degree of ethoxylation of about 2, for example, may comprise a significant amount of molecules which have no ethoxylate, 1 mole ethoxylate, 3 mole ethoxylate, and so on, while the distribution of ethoxylation can be broad, narrow or truncated, which still results in an overall weight average degree of ethoxylation of about 2. STS is particularly suitable for personal cleansing applications, and it has been a surprising and unexpected discovery of the present invention that when polyvinyl alcohol having a higher weight average molecular weight (e.g., from about 50,000 to about 400,000 Daltons, preferably from about 60,000 to about 300,000 Daltons, more preferably from about 70,000 to about 200,000 Daltons, most preferably from about 80,000 to about 150,000 Daltons) is used as the film-former and carrier, STS can be used as a major surfactant, i.e., present in an amount that is more than 50% by weight of the total surfactant content in the solid sheet, without adversely affecting the film-forming performance and stability of the overall composition. Correspondingly, in a particular embodiment of the present invention, STS is used as the major surfactant in the solid sheet. If present, the amount of STS in the solid sheet of the present invention may range from about 10% to about 70%, preferably from about 20% to about 65%, more preferably from about 40% to about 60%, by total weight of the solid sheet.

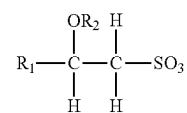
[0123] Another category of anionic surfactants suitable for practice of the present invention include alkyl sulfates. These materials have the respective formulae ROSO_3M , wherein R is alkyl or alkenyl of from about 6 to about 20 carbon atoms, x is 1 to 10, and M is a water-soluble cation such as ammonium, sodium, potassium and triethanolamine. Preferably, R has from about 6 to about 18, preferably from about 8 to about 16, more preferably from about 10 to about 14, carbon atoms. Previously, unalkoxyated $\text{C}_6\text{-C}_{20}$ linear or branched alkyl sulfates (AS) have been considered the preferred surfactants in dissolvable solid sheets, especially as the major surfactant therein, due to its compatibility with low molecular weight polyvinyl alcohols (e.g., those with a weight average molecular weight of no more than 50,000 Daltons) in film-forming performance and storage stability. However, it has been a surprising and unexpected discovery of the present invention that when polyvinyl alcohol having a higher weight average molecular weight (e.g., from about 50,000 to about 400,000 Daltons, preferably from about 60,000 to about 300,000 Daltons, more preferably from about 70,000 to about 200,000 Daltons, most preferably from about 80,000 to about 150,000 Daltons) is used as the film-former and carrier, other surfactants, such as LAS and/or STS, can be used as the major surfactant in the solid sheet, without adversely affecting the film-forming performance and stability of the overall composition. Therefore, in

a particularly preferred embodiment of the present invention, it is desirable to provide a solid sheet with no more than about 20%, preferably from 0% to about 10%, more preferably from 0% to about 5%, most preferably from 0% to about 1%, by weight of the solid sheet, of AS.

[0124] Another category of anionic surfactants suitable for practice of the present invention include $\text{C}_6\text{-C}_{20}$ linear or branched alkylalkoxy sulfates (AAS). Among this category, linear or branched alkylethoxy sulfates (AES) having the respective formulae $\text{RO}(\text{C}_2\text{H}_4\text{O})_x\text{SO}_3\text{M}$ are particularly preferred, wherein R is alkyl or alkenyl of from about 6 to about 20 carbon atoms, x is 1 to 10, and M is a water-soluble cation such as ammonium, sodium, potassium and triethanolamine. Preferably, R has from about 6 to about 18, preferably from about 8 to about 16, more preferably from about 10 to about 14, carbon atoms. The AES surfactants are typically made as condensation products of ethylene oxide and monohydric alcohol's having from about 6 to about 20 carbon atoms. Useful alcohols can be derived from fats, e.g., coconut oil or tallow, or can be synthetic. Lauryl alcohol and straight chain alcohol's derived from coconut oil are preferred herein. Such alcohol's are reacted with about 1 to about 10, preferably from about 3 to about 5, and especially about 3, molar proportions of ethylene oxide and the resulting mixture of molecular species having, for example, an average of 3 moles of ethylene oxide per mole of alcohol, is sulfated and neutralized. Highly preferred AES are those comprising a mixture of individual compounds, the mixture having an average alkyl chain length of from about 10 to about 16 carbon atoms and an average degree of ethoxylation of from about 1 to about 4 moles of ethylene oxide. If present, the amount of AAS in the solid sheet of the present invention may range from about 2% to about 40%, preferably from about 5% to about 30%, more preferably from about 8% to about 12%, by total weight of the solid sheet.

[0125] Other suitable anionic surfactants include water-soluble salts of the organic, sulfuric acid reaction products of the general formula $[\text{R}^1\text{-SO}_3\text{-M}]$, wherein R^1 is chosen from the group consisting of a straight or branched chain, saturated aliphatic hydrocarbon radical having from about 6 to about 20, preferably about 10 to about 18, carbon atoms; and M is a cation. Preferred are alkali metal and ammonium sulfonated C_{10-18} n-paraffins. Other suitable anionic surfactants include olefin sulfonates having about 12 to about 24 carbon atoms. The α -olefins from which the olefin sulfonates are derived are mono-olefins having about 12 to about 24 carbon atoms, preferably about 14 to about 16 carbon atoms. Preferably, they are straight chain olefins.

[0126] Another class of anionic surfactants suitable for use in the fabric and home care compositions is the β -alkyloxy alkane sulfonates. These compounds have the following formula:



where R_1 is a straight chain alkyl group having from about 6 to about 20 carbon atoms, R_2 is a lower alkyl group having from about 1 (preferred) to about 3 carbon atoms, and M is a water-soluble cation as hereinbefore described.

[0127] Additional examples of suitable anionic surfactants are the reaction products of fatty acids esterified with isethionic acid and neutralized with sodium hydroxide where, for example, the fatty acids are derived from coconut oil; sodium or potassium salts of fatty acid amides of methyl tauride in which the fatty acids, for example, are derived from coconut oil. Still other suitable anionic surfactants are the succinates, examples of which include disodium N-octadecylsulfosuccinamate; diammoniumlauryl sulfosuccinamate; tetrasodium N-(1,2-dicarboxyethyl)-N-octadecylsulfosuccinamate; diamyl ester of sodium sulfosuccinic acid; dihexyl ester of sodium sulfosuccinic acid; and dioctyl esters of sodium sulfosuccinic acid.

[0128] Nonionic surfactants that can be included into the solid sheet of the present invention may be any conventional nonionic surfactants, including but not limited to: alkyl alkoxyated alcohols, alkyl alkoxyated phenols, alkyl polysaccharides (especially alkyl glucosides and alkyl polyglucosides), polyhydroxy fatty acid amides, alkoxyated fatty acid esters, sucrose esters, sorbitan esters and alkoxyated derivatives of sorbitan esters, amine oxides, and the like. Preferred nonionic surfactants are those of the formula $R^1(OC_2H_4)_nOH$, wherein R^1 is a C_8 - C_{18} alkyl group or alkyl phenyl group, and n is from about 1 to about 80. Particularly preferred are C_8 - C_{18} alkyl ethoxyated alcohols having a weight average degree of ethoxylation from about 1 to about 20, preferably from about 5 to about 15, more preferably from about 7 to about 10, such as NEODOL® nonionic surfactants commercially available from Shell. Other non-limiting examples of nonionic surfactants useful herein include: C_6 - C_{12} alkyl phenol alkoxyates where the alkoxyate units may be ethyleneoxy units, propyleneoxy units, or a mixture thereof; C_{12} - C_{18} alcohol and C_6 - C_{12} alkyl phenol condensates with ethylene oxide/propylene oxide block polymers such as Pluronic® from BASF; C_{14} - C_{22} mid-chain branched alcohols (BA); C_{14} - C_{22} mid-chain branched alkyl alkoxyates, BAE_x, wherein x is from 1 to 30; alkyl polysaccharides, specifically alkyl polyglycosides; Polyhydroxy fatty acid amides; and ether capped poly(oxyalkylated) alcohol surfactants. Suitable nonionic surfactants also include those sold under the tradename Lutensol® from BASF.

[0129] In a preferred embodiment, the nonionic surfactant is selected from sorbitan esters and alkoxyated derivatives of sorbitan esters including sorbitan monolaurate (SPAN® 20), sorbitan monopalmitate (SPAN® 40), sorbitan monostearate (SPAN® 60), sorbitan tristearate (SPAN® 65), sorbitan monooleate (SPAN® 80), sorbitan trioleate (SPAN® 85), sorbitan isostearate, polyoxyethylene (20) sorbitan monolaurate (Tween® 20), polyoxyethylene (20) sorbitan monopalmitate (Tween® 40), polyoxyethylene (20) sorbitan monostearate (Tween® 60), polyoxyethylene (20) sorbitan monooleate (Tween® 80), polyoxyethylene (4) sorbitan monolaurate (Tween® 21), polyoxyethylene (4) sorbitan monostearate (Tween® 61), polyoxyethylene (5) sorbitan monooleate (Tween® 81), all available from Uniqema, and combinations thereof.

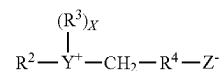
[0130] The most preferred nonionic surfactants for practice of the present invention include C_6 - C_{20} linear or branched alkylalkoxyated alcohols (AA) having a weight average degree of alkoxylation ranging from 5 to 15, more preferably C_{12} - C_{14} linear ethoxyated alcohols having a weight average degree of alkoxylation ranging from 7 to 9. If present, the amount of AA-type nonionic surfactant(s) in

the solid sheet of the present invention may range from about 2% to about 40%, preferably from about 5% to about 30%, more preferably from about 8% to about 12%, by total weight of the solid sheet.

[0131] Amphoteric surfactants suitable for use in the solid sheet of the present invention includes those that are broadly described as derivatives of aliphatic secondary and tertiary amines in which the aliphatic radical can be straight or branched chain and wherein one of the aliphatic substituents contains from about 8 to about 18 carbon atoms and one contains an anionic water solubilizing group, e.g., carboxy, sulfonate, sulfate, phosphate, or phosphonate. Examples of compounds falling within this definition are sodium 3-dodecyl-aminopropionate, sodium 3-dodecylaminopropane sulfonate, sodium lauryl sarcosinate, N-alkyltaurines such as the one prepared by reacting dodecylamine with sodium isethionate, and N-higher alkyl aspartic acids.

[0132] One category of amphoteric surfactants particularly suitable for incorporation into solid sheets with personal care applications (e.g., shampoo, facial or body cleanser, and the like) include alkylamphoacetates, such as lauroamphoacetate and cocoamphoacetate. Alkylamphoacetates can be comprised of monoacetates and diacetates. In some types of alkylamphoacetates, diacetates are impurities or unintended reaction products. If present, the amount of alkylamphoacetate(s) in the solid sheet of the present invention may range from about 2% to about 40%, preferably from about 5% to about 30%, more preferably from about 10% to about 20%, by total weight of the solid sheet.

[0133] Zwitterionic surfactants suitable include those that are broadly described as derivatives of aliphatic quaternary ammonium, phosphonium, and sulfonium compounds, in which the aliphatic radicals can be straight or branched chain, and wherein one of the aliphatic substituents contains from about 8 to about 18 carbon atoms and one contains an anionic group, e.g., carboxy, sulfonate, sulfate, phosphate, or phosphonate. Such suitable zwitterionic surfactants can be represented by the formula:



wherein R^2 contains an alkyl, alkenyl, or hydroxy alkyl radical of from about 8 to about 18 carbon atoms, from 0 to about 10 ethylene oxide moieties and from 0 to about 1 glyceryl moiety; Y is selected from the group consisting of nitrogen, phosphorus, and sulfur atoms; R^3 is an alkyl or monohydroxyalkyl group containing about 1 to about 3 carbon atoms; X is 1 when Y is a sulfur atom, and 2 when Y is a nitrogen or phosphorus atom; R^4 is an alkylene or hydroxyalkylene of from about 1 to about 4 carbon atoms and Z is a radical selected from the group consisting of carboxylate, sulfonate, sulfate, phosphonate, and phosphate groups.

[0134] Other zwitterionic surfactants suitable for use herein include betaines, including high alkyl betaines such as coco dimethyl carboxymethyl betaine, cocoamidopropyl betaine, cocobetaine, lauryl amidopropyl betaine, oleyl betaine, lauryl dimethyl carboxymethyl betaine, lauryl dimethyl alphacarboxyethyl betaine, cetyl dimethyl carboxymethyl betaine, lauryl bis-(2-hydroxyethyl) carboxymethyl betaine, stearyl bis-(2-hydroxypropyl) carboxymethyl

betaine, oleyl dimethyl gamma-carboxypropyl betaine, and lauryl bis-(2-hydroxypropyl)alpha-carboxyethyl betaine. The sulfobetaines may be represented by coco dimethyl sulfopropyl betaine, stearyl dimethyl sulfopropyl betaine, lauryl dimethyl sulfoethyl betaine, lauryl bis-(2-hydroxyethyl) sulfopropyl betaine and the like; amidobetaines and amidosulfobetaines, wherein the $\text{RCONH}(\text{CH}_2)_3$ radical, wherein R is a C_{11} - C_{17} alkyl, is attached to the nitrogen atom of the betaine are also useful in this invention.

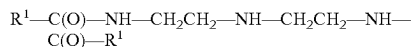
[0135] Cationic surfactants can also be utilized in the present invention, especially in fabric softener and hair conditioner products. When used in making products that contain cationic surfactants as the major surfactants, it is preferred that such cationic surfactants are present in an amount ranging from about 2% to about 30%, preferably from about 3% to about 20%, more preferably from about 5% to about 15% by total weight of the solid sheet.

[0136] Cationic surfactants may include DEQA compounds, which encompass a description of diamido actives as well as actives with mixed amido and ester linkages. Preferred DEQA compounds are typically made by reacting alkanolamines such as MDEA (methyldiethanolamine) and TEA (triethanolamine) with fatty acids. Some materials that typically result from such reactions include N,N-di(acyloxyethyl)-N,N-dimethylammonium chloride or N,N-di(acyloxyethyl)-N,N-methylhydroxyethylammonium methylsulfate wherein the acyl group is derived from animal fats, unsaturated, and polyunsaturated, fatty acids.

[0137] Other suitable actives for use as a cationic surfactant include reaction products of fatty acids with dialkylenetriamines in, e.g., a molecular ratio of about 2:1, the reaction products containing compounds of the formula:

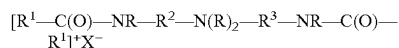


wherein R^1 , R^2 are defined as above, and each R^3 is a C_{1-6} alkylene group, preferably an ethylene group. Examples of these actives are reaction products of tallow acid, canola acid, or oleic acids with diethylenetriamine in a molecular ratio of about 2:1, the reaction product mixture containing N,N"-ditallowoyldiethylenetriamine, N,N"-dicanola-oyldiethylenetriamine, or N,N"-dioleoyldiethylenetriamine, respectively, with the formula:

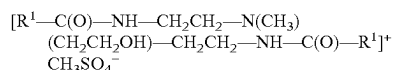


wherein R^2 and R^3 are divalent ethylene groups, R^1 is defined above and an acceptable examples of this structure when R^1 is the oleoyl group of a commercially available oleic acid derived from a vegetable or animal source, include EMERSOL®223LL or EMERSOL® 7021, available from Henkel Corporation.

[0138] Another active for use as a cationic surfactant has the formula:



wherein R, R^1 , R^2 , R^3 and X^- are defined as above. Examples of this active are the di-fatty amidoamines based softener having the formula:



wherein $\text{R}^1-\text{C}(\text{O})$ is an oleoyl group, soft tallow group, or a hardened tallow group available commercially from

Degussa under the trade names VARISOFT®222LT, VARISOFT® 222, and VARISOFT® 110, respectively.

[0139] A second type of DEQA ("DEQA (2)") compound suitable as a active for use as a cationic surfactant has the general formula:



wherein each Y, R, R^1 , and X^- have the same meanings as before. An example of a preferred DEQA (2) is the "propyl" ester quaternary ammonium fabric softener active having the formula 1,2-di(acyloxy)-3-trimethylammonio propane chloride.

[0140] Suitable polymeric surfactants for use in the personal care compositions of the present invention include, but are not limited to, block copolymers of ethylene oxide and fatty alkyl residues, block copolymers of ethylene oxide and propylene oxide, hydrophobically modified polyacrylates, hydrophobically modified celluloses, silicone polyethers, silicone copolyol esters, diquaternary polydimethylsiloxanes, and co-modified amino/polyether silicones.

[0141] In a preferred embodiment, the first surfactant may be selected from the group consisting of a C_6 - C_{20} linear alkylbenzene sulfonate (LAS), a C_6 - C_{20} linear or branched alkylalkoxy sulfates (AAS) having a weight average degree of alkoxylation ranging from 0.5 to 10, a C_6 - C_{20} linear or branched alkylalkoxylated alcohols (AA) having a weight average degree of alkoxylation ranging from 5 to 15, a C_6 - C_{20} linear or branched alkyl sulfates (AS) and any combinations thereof.

3. Plasticizers

[0142] In a preferred embodiment of the present invention, the flexible, porous, dissolvable solid sheet of the present invention may further comprise a plasticizer, preferably in the amount ranging from about 0.1% to about 25%, preferably from about 0.5% to about 20%, more preferably from about 1% to about 15%, most preferably from 2% to 12%, by total weight of the solid sheet. Correspondingly, the wet pre-mixture used for forming such solid sheet may comprise from about 0.02% to about 20% of a plasticizer by weight of the wet pre-mixture, in one embodiment from about 0.1% to about 10% of a plasticizer by weight of the wet pre-mixture, in one embodiment from about 0.5% to about 5% of a plasticizer by weight of the wet pre-mixture.

[0143] Suitable plasticizers for use in the present invention include, for example, polyols, copolyols, polycarboxylic acids, polyesters, dimethicone copolyols, and the like.

[0144] Examples of useful polyols include, but are not limited to: glycerin, diglycerin, ethylene glycol, polyethylene glycol (especially 200-600), propylene glycol, butylene glycol, pentylene glycol, glycerol derivatives (such as propoxylated glycerol), glycidol, cyclohexane dimethanol, hexanediol, 2,2,4-trimethylpentane-1,3-diol, pentaerythritol, urea, sugar alcohols (such as sorbitol, mannitol, lactitol, xylitol, maltitol, and other mono- and polyhydric alcohols), mono-, di- and oligo-saccharides (such as fructose, glucose, sucrose, maltose, lactose, high fructose corn syrup solids, and dextrans), ascorbic acid, sorbates, ethylene bisformamide, amino acids, and the like.

[0145] Examples of polycarboxylic acids include, but are not limited to citric acid, maleic acid, succinic acid, polyacrylic acid, and polymaleic acid.

[0146] Examples of suitable polyesters include, but are not limited to, glycerol triacetate, acetylated-monoglyceride,

diethyl phthalate, triethyl citrate, tributyl citrate, acetyl triethyl citrate, acetyl tributyl citrate.

[0147] Examples of suitable dimethicone copolyols include, but are not limited to, PEG-12 dimethicone, PEG/PPG-18/18 dimethicone, and PPG-12 dimethicone.

[0148] Other suitable plasticizers include, but are not limited to, alkyl and allyl phthalates; naphthalates; lactates (e.g., sodium, ammonium and potassium salts); sorbeth-30; urea; lactic acid; sodium pyrrolidone carboxylic acid (PCA); sodium hyaluronate or hyaluronic acid; soluble collagen; modified protein; monosodium L-glutamate; alpha & beta hydroxyl acids such as glycolic acid, lactic acid, citric acid, maleic acid and salicylic acid; glyceryl polymethacrylate; polymeric plasticizers such as polyquaterniums; proteins and amino acids such as glutamic acid, aspartic acid, and lysine; hydrogen starch hydrolysates; other low molecular weight esters (e.g., esters of C₂-C₁₀ alcohols and acids); and any other water soluble plasticizer known to one skilled in the art of the foods and plastics industries; and mixtures thereof.

[0149] Particularly preferred examples of plasticizers include glycerin, ethylene glycol, polyethylene glycol, propylene glycol, and mixtures thereof. Most preferred plasticizer is glycerin.

4. Additional Ingredients

[0150] In addition to the above-described ingredients, e.g., the water-soluble polymer, the surfactant(s) and the plasticizer, the solid sheet of the present invention may comprise one or more additional ingredients, depending on its intended application. Such one or more additional ingredients may be selected from the group consisting of fabric care actives, dishwashing actives, hard surface cleaning actives, beauty and/or skin care actives, personal cleansing actives, hair care actives, oral care actives, feminine care actives, baby care actives, a bittering agent and any combinations thereof. In a preferred embodiment, the solid sheet of the present invention may comprise a bittering agent.

[0151] Suitable fabric care actives include but are not limited to: organic solvents (linear or branched lower C₁-C₈ alcohols, diols, glycerols or glycols; lower amine solvents such as C₁-C₄ alkanolamines, and mixtures thereof; more specifically 1,2-propanediol, ethanol, glycerol, monoethanolamine and triethanolamine), carriers, hydrotropes, builders, chelants, dispersants, enzymes and enzyme stabilizers, catalytic materials, bleaches (including photobleaches) and bleach activators, perfumes (including encapsulated perfumes or perfume microcapsules), colorants (such as pigments and dyes, including hueing dyes), brighteners, dye transfer inhibiting agents, clay soil removal/anti-redeposition agents, structurants, rheology modifiers, suds suppressors, processing aids, fabric softeners, anti-microbial agents, and the like.

[0152] Suitable hair care actives include but are not limited to: moisture control materials of class II for frizz reduction (salicylic acids and derivatives, organic alcohols, and esters), cationic surfactants (especially the water-insoluble type having a solubility in water at 25° C. of preferably below 0.5 g/100 g of water, more preferably below 0.3 g/100 g of water), high melting point fatty compounds (e.g., fatty alcohols, fatty acids, and mixtures thereof with a melting point of 25° C. or higher, preferably 40° C. or higher, more preferably 45° C. or higher, still more preferably 50° C. or higher), silicone compounds, condi-

tioning agents (such as hydrolyzed collagen with tradename Peptin 2000 available from Hormel, vitamin E with tradename Emix-d available from Eisai, panthenol available from Roche, panthenyl ethyl ether available from Roche, hydrolyzed keratin, proteins, plant extracts, and nutrients), preservatives (such as benzyl alcohol, methyl paraben, propyl paraben and imidazolidinyl urea), pH adjusting agents (such as citric acid, sodium citrate, succinic acid, phosphoric acid, sodium hydroxide, sodium carbonate), salts (such as potassium acetate and sodium chloride), coloring agents, perfumes or fragrances, sequestering agents (such as disodium ethylenediamine tetra-acetate), ultraviolet and infrared screening and absorbing agents (such as octyl salicylate), hair bleaching agents, hair perming agents, hair fixatives, anti-dandruff agents, anti-microbial agents, hair growth or restorer agents, co-solvents or other additional solvents, and the like.

[0153] Suitable beauty and/or skin care actives include those materials approved for use in cosmetics and that are described in reference books such as the CTEA Cosmetic Ingredient Handbook, Second Edition, The Cosmetic, Toiletries, and Fragrance Association, Inc. 1988, 1992. Further non-limiting examples of suitable beauty and/or skin care actives include preservatives, perfumes or fragrances, coloring agents or dyes, thickeners, moisturizers, emollients, pharmaceutical actives, vitamins or nutrients, sunscreens, deodorants, sensates, plant extracts, nutrients, astringents, cosmetic particles, absorbent particles, fibers, anti-inflammatory agents, skin lightening agents, skin tone agent (which functions to improve the overall skin tone, and may include vitamin B3 compounds, sugar amines, hexamidine compounds, salicylic acid, 1,3-dihydroxy-4-alkylbenzene such as hexylresorcinol and retinoids), skin tanning agents, exfoliating agents, humectants, enzymes, antioxidants, free radical scavengers, anti-wrinkle actives, anti-acne agents, acids, bases, minerals, suspending agents, pH modifiers, pigment particles, anti-microbial agents, insect repellents, shaving lotion agents, co-solvents or other additional solvents, and the like.

[0154] Suitable bittering agent include a denatonium salt or a derivative thereof. In one aspect, the bittering agent is a denatonium salt selected from the group consisting of denatonium chloride, denatonium citrate, denatonium saccharide, denatonium carbonate, denatonium acetate, denatonium benzoate, and mixtures thereof. In one aspect, the solid sheet comprises a first denatonium salt and the coating composition comprises a second denatonium salt that is different than the first denatonium salt.

[0155] A particularly preferred bittering agent is denatonium benzoate, also known as phenylmethyl-[2-[(2,6-dimethylphenyl)amino]-2-oxoethyl]-diethylammonium benzoate, CAS no. 3734-33-6. Denatonium benzoate is commercially sold as BITREX®, available from Macfarlan Smith, Edinburgh, Scotland, UK.

[0156] In some aspects, the bittering agent is a natural bitter substance. In some aspects, the bittering agent has a bitter value of from about 1000 to about 200000. In some aspects, the bittering agent is a natural bitter substance with a bitter value of from about 1000 to about 200000, where the natural bitter substance is selected from the group consisting of glycosides, isoprenoids, alkaloids, amino acids, and mixtures thereof. For example, suitable bittering agents also include Quercetin (3,3',4',5,7-pentahydroxyflavone); Naringin (4',5,7-Trihydroxyflavanone-7-rhamnoglucoside);

Aucubin; Amarogentin; Dihydrofoliamentin; Gentiopicroside; Gentiopicrin; Swertiamarin; Swerosid; Gentioflavosid; Centaurosid; Methiafolin; Harpagoside; Centapikrin; Sailicin; Kondurangin; Absinthin; Artabsin; Cnicin; Lactucin; Lactucopicrin; Salomonitenolid; α -thujone; β -thujone; Desoxy Limonene; Limonin; Ichangin; iso-Obacunoic Acid; Obacunone; Obacunoic Acid; Nomilin; Ichangin; Nomilinoic acid; Marrubin; Pramarrubin; Carnosol; Carnosic acid; Quassin; Quinine hydrochloride; Quinine sulfate; Quinine dihydrochloride; Columbine; Caffeine; Threonine; Methionine; Phenylalanine; Tryptophan; Arginine; Histidine; Valine; Aspartic acid; Sucrose octaacetate; and mixtures thereof. Other suitable bittering agents include quinine bisulfate and hop extract (e.g., humulone).

[0157] The solid sheet may comprise from about 0.00001% to about 1%, or about 0.0001% to about 0.5%, or about 0.001% to about 0.25%, or about from about 0.01% to about 0.1%, by weight of the solid sheet, of a bittering agent. In some aspects, the solid sheet comprises a bittering agent in a sufficient amount to provide a bitter taste.

[0158] The solid sheet of the present invention may further comprise other optional ingredients that are known for use or otherwise useful in compositions, provided that such optional materials are compatible with the selected essential materials described herein, or do not otherwise unduly impair product performance.

[0159] Non-limiting examples of product type embodiments that can be formed by the solid sheet of the present invention include laundry detergent products, fabric softening products, hand cleansing products, hair shampoo or other hair treatment products, body cleansing products, shaving preparation products, dish cleaning products, personal care substrates containing pharmaceutical or other skin care actives, moisturizing products, sunscreen products, beauty or skin care products, deodorizing products, oral care products, feminine cleansing products, baby care products, fragrance-containing products, and so forth.

VI. Formulations of Coating Composition

[0160] The coating composition (also referred to as “juice” herein) according to the present disclosure may comprise a second surfactant. The coating composition may have a viscosity of from about 1 cps to about 25,000 cps, preferably from about 2 cps to about 10,000 cps, more preferably from about 3 cps to about 5,000 cps, most preferably from about 1,000 cps to about 5,000 cps, as measured at about 20° C. and 1 s⁻¹. The viscosity values are measured using a Malvern Kinexus Lab+ rheometer with cone and plate geometry (CP1/50 SR3468 SS), a gap width of 0.054 mm, a temperature of 20° C. and a shear rate of 1.0 reciprocal seconds for a period of 360 seconds.

1. Second Surfactants

[0161] The second surfactant may function as active ingredients for delivering a desired cleansing benefit. In some embodiments, the second surfactant may be the same with the first surfactant. In other embodiments, the second surfactant may be different from the first surfactant. The second surfactant may be selected from the group consisting of anionic surfactants, nonionic surfactants, cationic surfactants, zwitterionic surfactants, amphoteric surfactants, polymeric surfactants and any combinations thereof. Depending on the desired application of the solid article and the desired

consumer benefit to be achieved, different surfactants can be selected. In a preferred embodiment of the present invention, the second surfactant may comprise a non-ionic surfactant.

[0162] Any anionic surfactants, nonionic surfactants, cationic surfactants, zwitterionic surfactants, amphoteric surfactants, and/or polymeric surfactants listed in the section of “FIRST SURFACTANTS” or any combinations thereof may be used as the second surfactant.

[0163] In a preferred embodiment, the second surfactant may be selected from the group consisting of a C₆-C₂₀ linear alkylbenzene sulfonate (LAS), a C₆-C₂₀ linear or branched alkylalkoxy sulfates (AAS) having a weight average degree of alkoxylation ranging from 0.5 to 10, a C₆-C₂₀ linear or branched alkylalkoxylated alcohols (AA) having a weight average degree of alkoxylation ranging from 5 to 15, a C₆-C₂₀ linear or branched alkyl sulfates (AS) and any combinations thereof; and

[0164] In a more preferred embodiment, the second surfactant may comprise a C₆-C₂₀ linear or branched alkylalkoxylated alcohols (AA) having a weight average degree of alkoxylation ranging from 5 to 15, preferably C₁₂-C₁₄ linear ethoxylated alcohols having a weight average degree of alkoxylation ranging from 7 to 9.

[0165] Particularly, the coating composition may comprise from 1% to 95%, preferably from 1% to 85%, more preferably from 10% to 80%, for example 1%, 2%, 3%, 4%, 5%, 8%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or any ranges therebetween, of the second surfactant by total weight of the coating composition.

2. Solvent

[0166] The coating composition may further comprise a solvent that may be preferably selected from the group consisting of glycerol, propylene glycol, 1,3-propanediol, diethylene glycol, dipropylene glycol, ethanolamine, ethanol, water and any combinations thereof. The solvent may be an organic solvent.

[0167] Particularly, the solvent may be selected from the group consisting of glycerol, diethylene glycol, dipropylene glycol, ethanol, water and any combinations thereof. More particularly, the solvent may be dipropylene glycol. The presence of the solvent in the coating composition may bring about an even more improved dissolution profile.

[0168] Particularly, the coating composition may comprise from 0.1% to 99%, preferably from 1% to 70%, more preferably from 2% to 30%, for example 1%, 2%, 3%, 4%, 5%, 8%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or any ranges therebetween, of the solvent by total weight of the coating composition.

[0169] Preferably, the coating composition may comprise less than 30%, preferably less than 25%, more preferably less than 20%, yet more preferably less than 10%, yet more preferably less than 5%, yet more preferably less than 3%, yet more preferably less than 1%, most preferably less than 0.5%, for example 0.01%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 1%, 2%, 3%, 4%, 5% or any ranges therebetween, of water by total weight of the coating composition. The presence of a appropriate amount of water may bring about additional benefit, for example to facilitate the attachment between sheets and/or to modify rheology and/or even to further improve the dissolution.

3. Rheology Modifier

[0170] The coating composition may further comprise a rheology modifier that is preferably selected from the group consisting of cellulose and derivatives; a guar and guar derivatives; polyethylene oxide, polypropylene oxide, and POE-PPO copolymers; polyvinylpyrrolidone, crosslinked polyvinylpyrrolidone and derivatives; polyvinylalcohol and derivatives; polyethyleneimine and derivatives; finely divided inorganic particles such as sodium carbonate and sodium sulphate; silicon dioxide; water-swellaable clays; and gums. The presence of the rheology modifier may facilitate to modify the rheology, for example the viscosity.

[0171] Particularly, The coating composition may comprise from 0.1% to 95%, preferably from 0.5% to 85%, more preferably from 1% to 50%, for example 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or any ranges therebetween, of the rheology modifier by total weight of the coating composition.

[0172] In a further embodiment, the rheology modifier may be cellulose and derivatives, in which non-limiting examples include microcrystalline cellulose, carboxymethylcelluloses, hydroxyethylcellulose, hydroxypropylcellulose, hydroxypropylmethylcellulose, methylcellulose, ethylcellulose, nitro cellulose, cellulose sulfate, cellulose powder, and hydrophobically modified celluloses

[0173] In an embodiment, the rheology modifier may be a guar and guar derivatives, in which non-limiting examples include hydroxypropyl guar, and hydroxypropyl guar hydroxypropyl trimonium chloride.

[0174] In an embodiment, the rheology modifier may be polyethylene oxide, polypropylene oxide, and POE-PPO copolymers.

[0175] In an embodiment, the rheology modifier may be polyvinylpyrrolidone, crosslinked polyvinylpyrrolidone and derivatives.

[0176] In a further embodiment, the rheology modifier may be polyvinylalcohol and derivatives.

[0177] In a further embodiment, the rheology modifier may be polyethyleneimine and derivatives.

[0178] In another embodiment, the rheology modifier may be silicon dioxide, in which nonlimiting examples include fumed silica, precipitated silica, and silicone-surface treated silica.

[0179] In an embodiment, the rheology modifier may be water-swellaable clays, in which non-limiting examples include laponite, bentolite, montmorillonite, smectite, and hectonite.

[0180] In an embodiment, the rheology modifier may be gums, in which non-limiting examples include xanthan gum, guar gum, hydroxypropyl guar gum, Arabia gum, tragacanth, galactan, carob gum, karaya gum, and locust bean gum.

4. Perfume

[0181] The coating composition may further comprise a perfume (for example, free perfumes, encapsulated perfumes). Preferably, the perfume may be free perfumes, perfume microcapsules, or any combinations thereof. Particularly, the coating composition may comprise from 1% to 99%, preferably from 5% to 90%, more preferably from 10% to 80%, for example 1%, 2%, 3%, 4%, 5%, 8%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%,

65%, 70%, 75%, 80%, 85%, 90%, 95% or any ranges therebetween, of the perfume by total weight of the coating composition.

[0182] In some embodiments, the ratio by weight of said second surfactant to said perfume in the coating composition may be from about 1:50 to about 50:1, preferably from about 1:20 to about 20:1, more preferably from about 1:1 to about 10:1, most preferably from about 2:1 to about 8:1, for example 1:10, 1:9, 1:8, 1:7, 1:6, 1:5, 1:4, 1:3, 1:2, 1:1, 2:1, 3:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 15:1, 20:1 or any ranges therebetween.

[0183] In some embodiments, at least 50%, preferably at least 70%, more preferably at least 90%, most preferably at least 99%, of perfume in the solid article according to the present disclosure is present in the coating composition. It may bring about improved performance of perfumes, for example longevity, perfume stability, deposition or release benefit.

5. Additional Ingredients

[0184] In addition to the above-described ingredients, the coating composition of the present invention may comprise one or more additional ingredients, depending on its intended application. Such one or more additional ingredients may be selected from the group consisting of fabric care actives, dishwashing actives, hard surface cleaning actives, beauty and/or skin care actives, personal cleansing actives, hair care actives, oral care actives, feminine care actives, baby care actives, a bittering agent and any combinations thereof.

[0185] Particularly, the coating composition may further comprise an additional ingredient selected from the group consisting of a silicone, a softening agent, a bleaching agent, an enzyme, an anti-bac agent, an anti-oxidant, a brightener, a hueing dye, a polymer, a personal care active (for example, emollients, humectants, and conditioning agents) and any combinations thereof.

[0186] The coating composition may comprise from 0.0001% to 99%, preferably from 1% to 95%, more preferably from 10% to 80%, for example 0.001%, 0.01%, 0.1%, 1%, 2%, 3%, 4%, 5%, 8%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or any ranges therebetween, of the additional ingredient by total weight of the coating composition.

VII. Conversion of Multiple Solid Sheets and Coating Composition into Multilayer Dissolvable Solid Articles Containing Coating Composition

[0187] Once the flexible, dissolvable, porous solid sheet of the present invention is formed, as described hereinabove, two or more of such sheets can be further combined and/or treated by applying the coating composition to form dissolvable solid articles of any desirable three-dimensional shapes, including but not limited to: spherical, cubic, rectangular, oblong, cylindrical, rod, sheet, flower-shaped, fan-shaped, star-shaped, disc-shaped, and the like. The sheets can be combined and/or treated by any means known in the art, examples of which include but are not limited to, chemical means, mechanical means, and combinations thereof. Such combination and/or treatment steps are hereby collectively referred to as a "conversion" process, i.e., which functions to convert two or more flexible, dissolvable, porous sheets of the present invention into a dissolvable solid article containing a coating composition.

[0188] It has been a surprising and unexpected discovery of the present invention that three-dimensional multilayer solid articles containing the coating composition have significantly improved dissolution profiles than multilayer solid articles having the same amount of overall surfactants but without the coating composition.

[0189] Furthermore, the multilayer dissolvable solid articles of the present invention may be characterized by a maximum dimension D and a minimum dimension z (which is perpendicular to the maximum dimension), while the ratio of D/z (hereinafter also referred to as the "Aspect Ratio") ranges from 1 to about 10, preferably from about 1.4 to about 9, preferably from about 1.5 to about 8, more preferably from about 2 to about 7. Note that when the Aspect Ratio is 1, the dissolvable solid article has a spherical shape. When the Aspect Ratio is about 1.4, the dissolvable solid article has a cubical shape. The multilayer dissolvable solid article of the present invention may have a minimal dimension z that is greater than about 3 mm but less than about 20 cm, preferably from about 4 mm to about 10 cm, more preferably from about 5 mm to about 30 mm.

[0190] The above-described multilayer dissolvable solid article may comprise more than two of such flexible, dissolvable, porous sheets. For example, it may comprise from about 4 to about 50, preferably from about 5 to about 40, more preferably from about 6 to about 30, of the flexible, dissolvable, porous sheets. The improved OCF structures in the flexible, dissolvable, porous sheets made according to the present invention allow stacking of many sheets (e.g., 15-40) together, while still providing a satisfactory overall dissolution rate for the stack.

[0191] In a particularly preferred embodiment of the present invention, the multilayer dissolvable solid article comprises from 15 to 40 layers of the above-described flexible, dissolvable, porous sheets and has an aspect ratio ranging from about 2 to about 7.

[0192] Particularly, the coating composition may be applied between individual sheets of the multilayer dissolvable solid article by any appropriate means, e.g., by spraying, sprinkling, dusting, coating, spreading, dipping, injecting, rolling, or even vapor deposition. More particularly, the coating composition may be applied on one or both of contacting surfaces of adjacent sheets in the stack. In a preferred embodiment, in order to avoid interference of the coating composition with the cutting seal or edge seal near the peripherals of the individual sheets, the coating composition may be applied in a central region on each of the applied surfaces of the respective sheets, which is preferably defined as a region that is spaced apart from the peripherals of such adjacent sheets by a distance that is at least 5%, preferably at least 10%, more preferably at least 15%, most preferably at least 20%, of the maximum Dimension D. In an alternative preferred embodiment, said coating composition is applied throughout the applied surfaces of the respective sheets, preferably wherein the applied area accounts for at least 90%, preferably 95%, more preferably 98%, most preferably 99% of the total area of the applied surfaces.

[0193] In a preferred embodiment, the coating composition may be applied on one or both contacting surfaces of any adjacent sheets in the solid article. In another preferred embodiment, the coating composition may be applied on one or both contacting surfaces of middle two sheets in the stack. In yet another preferred embodiment, the coating

composition may be applied on one or both of contacting surfaces of any two adjacent sheets in the stack excluding the two outermost sheets.

[0194] The term of "middle two sheets" as used herein means the two adjacent sheets that are located in the middle of the sequence of sheets stacked together. Particularly, if the total number of sheets is an odd number (e.g., 7), middle two sheets include the sheet that is located in the middle and any of two adjacent sheets thereof (e.g., the 3rd and 4th sheets or the 4th and 5th sheets); and if the total number of sheets is an even number (e.g., 6), middle two sheets include the two sheets that are located in the middle (e.g., the 3rd and 4th sheets).

[0195] The multilayer dissolvable solid article may comprise from 0.1% to 90%, preferably from 1% to 80%, more preferably from 5% to 70%, and most preferably from 10% to 60%, for example 1%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90% or any ranges therebetween, of the coating composition by total weight of the article.

[0196] The multilayer dissolvable solid article may comprise from 10% to 99.9%, preferably from 20% to 99%, more preferably from 30% to 95%, and most preferably from 40% to 90%, for example 1%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any ranges therebetween, of the solid sheet by total weight of the article.

[0197] The multilayer dissolvable solid article of the present invention may comprise individual sheets of different colors, which are visual from an external surface (e.g., one or more side surfaces) of such article. Such visible sheets of different colors are aesthetically pleasing to the consumers. Further, the different colors of individual sheets may provide visual cues indicative of different benefit agents contained in the individual sheets. For example, the multilayer dissolvable solid article may comprise a first sheet that has a first color and contains a first benefit agent and a second sheet that has a second color and contains a second benefit, while the first color provides a visual cue indicative of the first benefit agent, and while the second color provides a visual cue indicative of the second benefit agent.

Test Methods

Test 1: Scanning Electron Microscopic (SEM) Method for Determining Surface Average Pore Diameter of the Sheet Article

[0198] A Hitachi TM3000 Tabletop Microscope (S/N: 123104-04) is used to acquire SEM micrographs of samples. Samples of the solid sheet articles of the present invention are approximately 1 cm×1 cm in area and cut from larger sheets. Images are collected at a magnification of 50×, and the unit is operated at 15 kV. A minimum of 5 micrograph images are collected from randomly chosen locations across each sample, resulting in a total analyzed area of approximately 43.0 mm² across which the average pore diameter is estimated.

[0199] The SEM micrographs are then firstly processed using the image analysis toolbox in Matlab. Where required, the images are converted to grayscale. For a given image, a histogram of the intensity values of every single pixel is generated using the 'imhist' Matlab function. Typically, from such a histogram, two separate distributions are obvious, corresponding to pixels of the brighter sheet surface and

pixels of the darker regions within the pores. A threshold value is chosen, corresponding to an intensity value between the peak value of these two distributions. All pixels having an intensity value lower than this threshold value are then set to an intensity value of 0, while pixels having an intensity value higher are set to 1, thus producing a binary black and white image. The binary image is then analyzed using ImageJ (<https://imagej.nih.gov>, version 1.52a), to examine both the pore area fraction and pore size distribution. The scale bar of each image is used to provide a pixel/mm scaling factor. For the analysis, the automatic thresholding and the analyze particles functions are used to isolate each pore. Output from the analyze function includes the area fraction for the overall image and the pore area and pore perimeter for each individual pore detected.

[0200] Average Pore Diameter is defined as D_{A50} : 50% of the total pore area is comprised of pores having equal or smaller hydraulic diameters than the D_{A50} average diameter.

$$\text{Hydraulic diameter} = \frac{4 * \text{Pore area (m}^2\text{)}}{\text{Pore perimeter (m)}}$$

[0201] It is an equivalent diameter calculated to account for the pores not all being circular.

Test 2: Micro-Computed Tomographic (uCT) Method for Determining Overall or Regional Average Pore Size and Average Cell Wall Thickness of the Open Cell Foams (OCF)

[0202] Porosity is the ratio between void-space to the total space occupied by the OCF. Porosity can be calculated from pCT scans by segmenting the void space via thresholding and determining the ratio of void voxels to total voxels. Similarly, solid volume fraction (SVF) is the ratio between solid-space to the total space, and SVF can be calculated as the ratio of occupied voxels to total voxels. Both Porosity and SVF are average scalar-values that do not provide structural information, such as, pore size distribution in the height-direction of the OCF, or the average cell wall thickness of OCF struts.

[0203] To characterize the 3D structure of the OCFs, samples are imaged using a μ CT X-ray scanning instrument capable of acquiring a dataset at high isotropic spatial resolution. One example of suitable instrumentation is the SCANCO system model 50 μ CT scanner (Scanco Medical AG, Brüttisellen, Switzerland) operated with the following settings: energy level of 45 kVp at 133 μ A; 3000 projections; 15 mm field of view; 750 ms integration time; an averaging of 5; and a voxel size of 3 μ m per pixel. After scanning and subsequent data reconstruction is complete, the scanner system creates a 16 bit data set, referred to as an ISQ file, where grey levels reflect changes in x-ray attenuation, which in turn relates to material density. The ISQ file is then converted to 8 bit using a scaling factor.

[0204] Scanned OCF samples are normally prepared by punching a core of approximately 14 mm in diameter. The OCF punch is laid flat on a low-attenuating foam and then mounted in a 15 mm diameter plastic cylindrical tube for scanning. Scans of the samples are acquired such that the entire volume of all the mounted cut sample is included in the dataset. From this larger dataset, a smaller sub-volume of the sample dataset is extracted from the total cross section of the scanned OCF, creating a 3D slab of data, where pores can be qualitatively assessed without edge/boundary effects.

[0205] To characterize pore-size distribution in the height-direction, and the strut-size, Local Thickness Map algorithm, or LTM, is implemented on the subvolume dataset. The LTM Method starts with a Euclidean Distance Mapping

(EDM) which assigns grey level values equal to the distance each void voxel is from its nearest boundary. Based on the EDM data, the 3D void space representing pores (or the 3D solid space representing struts) is tessellated with spheres sized to match the EDM values. Voxels enclosed by the spheres are assigned the radius value of the largest sphere. In other words, each void voxel (or solid voxel for struts) is assigned the radial value of the largest sphere that both fits within the void space boundary (or solid space boundary for struts) and includes the assigned voxel.

[0206] The 3D labelled sphere distribution output from the LTM data scan can be treated as a stack of two dimensional images in the height-direction (or Z-direction) and used to estimate the change in sphere diameter from slice to slice as a function of OCF depth. The strut thickness is treated as a 3D dataset and an average value can be assessed for the whole or parts of the subvolume. The calculations and measurements were done using AVIZO Lite (9.2.0) from Thermo Fisher Scientific and MATLAB (R2017a) from Mathworks.

Test 3: Percent Open Cell Content of the Sheet Article

[0207] The Percent Open Cell Content is measured via gas pycnometry. Gas pycnometry is a common analytical technique that uses a gas displacement method to measure volume accurately. Inert gases, such as helium or nitrogen, are used as the displacement medium. A sample of the solid sheet article of the present invention is sealed in the instrument compartment of known volume, the appropriate inert gas is admitted, and then expanded into another precision internal volume. The pressure before and after expansion is measured and used to compute the sample article volume.

[0208] ASTM Standard Test Method D2856 provides a procedure for determining the percentage of open cells using an older model of an air comparison pycnometer. This device is no longer manufactured. However, one can determine the percentage of open cells conveniently and with precision by performing a test which uses Micromeritics' AccuPyc Pycnometer. The ASTM procedure D2856 describes 5 methods (A, B, C, D, and E) for determining the percent of open cells of foam materials. For these experiments, the samples can be analyzed using an Accupyc 1340 using nitrogen gas with the ASTM foampyc software. Method C of the ASTM procedure is to be used to calculate to percent open cells. This method simply compares the geometric volume as determined using calipers and standard volume calculations to the open cell volume as measured by the Accupyc, according to the following equation:

$$\text{Open cell percentage} = \frac{\text{Open cell volume of sample}}{\text{Geometric volume of sample}} * 100$$

[0209] It is recommended that these measurements be conducted by Micromeritics Analytical Services, Inc. (One Micromeritics Dr, Suite 200, Norcross, GA 30093). More information on this technique is available on the Micromeritics Analytical Services web sites (www.particletesting.com or www.micromeritics.com), or published in "Analytical Methods in Fine particle Technology" by Clyde Orr and Paul Webb.

Test 4: Final Moisture Content of the Sheet Article

[0210] Final moisture content of the solid sheet article of the present invention is obtained by using a Mettler Toledo HX204 Moisture Analyzer (S/N B706673091). A minimum

of 1 g of the dried sheet article is placed on the measuring tray. The standard program is then executed, with additional program settings of 10 minutes analysis time and a temperature of 110° C.

Test 5: Thickness of the Sheet Article

[0211] Thickness of the flexible, porous, dissolvable solid sheet article of the present invention is obtained by using a micrometer or thickness gage, such as the Mitutoyo Corporation Digital Disk Stand Micrometer Model Number IDS-1012E (Mitutoyo Corporation, 965 Corporate Blvd, Aurora, IL, USA 60504). The micrometer has a 1-inch diameter platen weighing about 32 grams, which measures thickness at an application pressure of about 0.09 psi (6.32 gm/cm²).

[0212] The thickness of the flexible, porous, dissolvable solid sheet article is measured by raising the platen, placing a section of the sheet article on the stand beneath the platen, carefully lowering the platen to contact the sheet article, releasing the platen, and measuring the thickness of the sheet article in millimeters on the digital readout. The sheet article should be fully extended to all edges of the platen to make sure thickness is measured at the lowest possible surface pressure, except for the case of more rigid substrates which are not flat.

Test 6: Basis Weight of the Sheet Article

[0213] Basis Weight of the flexible, porous, dissolvable solid sheet article of the present invention is calculated as the weight of the sheet article per area thereof (grams/m²). The area is calculated as the projected area onto a flat surface perpendicular to the outer edges of the sheet article. The solid sheet articles of the present invention are cut into sample squares of 10 cm×10 cm, so the area is known. Each of such sample squares is then weighed, and the resulting weight is then divided by the known area of 100 cm² to determine the corresponding basis weight.

[0214] For an article of an irregular shape, if it is a flat object, the area is thus computed based on the area enclosed within the outer perimeter of such object. For a spherical object, the area is thus computed based on the average diameter as $3.14 \times (\text{diameter}/2)^2$. For a cylindrical object, the area is thus computed based on the average diameter and average length as diameter×length. For an irregularly shaped three-dimensional object, the area is computed based on the side with the largest outer dimensions projected onto a flat surface oriented perpendicularly to this side. This can be accomplished by carefully tracing the outer dimensions of the object onto a piece of graph paper with a pencil and then computing the area by approximate counting of the squares and multiplying by the known area of the squares or by taking a picture of the traced area (shaded-in for contrast) including a scale and using image analysis techniques.

Test 7: Density of the Sheet Article

[0215] Density of the flexible, porous, dissolvable solid sheet article of the present invention is determined by the equation: Calculated Density=Basis Weight of porous solid/

(Porous Solid Thickness×1,000). The Basis Weight and Thickness of the dissolvable porous solid are determined in accordance with the methodologies described hereinabove.

Test 8: Specific Surface Area of the Sheet Article

[0216] The Specific Surface Area of the flexible, porous, dissolvable solid sheet article is measured via a gas adsorption technique. Surface Area is a measure of the exposed surface of a solid sample on the molecular scale. The BET (Brunauer, Emmet, and Teller) theory is the most popular model used to determine the surface area and is based upon gas adsorption isotherms. Gas Adsorption uses physical adsorption and capillary condensation to measure a gas adsorption isotherm. The technique is summarized by the following steps; a sample is placed in a sample tube and is heated under vacuum or flowing gas to remove contamination on the surface of the sample. The sample weight is obtained by subtracting the empty sample tube weight from the combined weight of the degassed sample and the sample tube. The sample tube is then placed on the analysis port and the analysis is started. The first step in the analysis process is to evacuate the sample tube, followed by a measurement of the free space volume in the sample tube using helium gas at liquid nitrogen temperatures. The sample is then evacuated a second time to remove the helium gas. The instrument then begins collecting the adsorption isotherm by dosing krypton gas at user specified intervals until the requested pressure measurements are achieved. Samples may then analyzed using an ASAP 2420 with krypton gas adsorption. It is recommended that these measurements be conducted by Micromeritics Analytical Services, Inc. (One Micromeritics Dr, Suite 200, Norcross, GA 30093). More information on this technique is available on the Micromeritics Analytical Services web sites (www.particletesting.com or www.micromeritics.com), or published in a book, "Analytical Methods in Fine Particle Technology", by Clyde Orr and Paul Webb.

Test 9: Dissolution Rate of the Sheet Article

[0217] The dissolution rate of dissolvable sheets or solid articles of the present invention is measured as follows:

[0218] 1. 400 ml of deionized water at room temperature (25° C.) is added to a 1 L beaker, and the beaker is then placed on a magnetic stirrer plate.

[0219] 2. A magnetic stirrer bar having length 23 mm and thickness of 10 mm is placed in the water and set to rotate at 300 rpm.

[0220] 3. A Mettler Toledo S230 conductivity meter is calibrated to 1413 μS/cm and the probe placed in the beaker of water.

[0221] 4. For each experiment, the number of samples is chosen such that a minimum of 0.2 g of sample is dissolved in the water.

[0222] 5. The data recording function on the conductivity meter is started and the samples are dropped into the beaker. For 5 seconds a flat steel plate with diameter similar to that of the glass beaker is used to submerge the samples below the surface of the water and prevent them from floating to the surface.

[0223] 6. The conductivity is recorded for at least 10 minutes, until a steady state value is reached.

[0224] 7. In order to calculate the time required to reach 95% dissolution, a 10 second moving average is firstly

calculated from the conductivity data. The time at which this moving average surpassed 95% of the final steady state conductivity value is then estimated and taken as the time required to achieve 95% dissolution.

Test 10: Gelling Test of the Sheet Article

[0225] The foam gelling during the dissolution of dissolvable, solid sheets or articles is measured by rheometer oscillatory testing as follows:

[0226] A Malvern Kinexus Lab+ rheometer with 40 mm flat plate stainless steel spindle (PU40 SR4067 SS) and flat stainless steel baseplate (PLS61 S2837 SS) is used. The standard oscillation test “Measure 0033 Single frequency strain controlled” is loaded in the Kinexus software with the following parameters: 1 Hz oscillation frequency, 1% strain, 10 minutes total measurement time. The test program is then modified by deleting the temperature setpoint step from the software, and adding a ‘set gap’ step, with the gap setpoint at 1 mm. The data recording frequency is set to 1 data point per second. These modifications are made so that once initiated, the program would immediately go to the setpoint gap and begin measurement as soon as the setpoint gap was reached, and the foam and water come into contact. Prior to starting the experiment, the base plate temperature was set to a constant value setpoint of 20° C.

[0227] With the spindle inserted into the rheometer, and after running the calibration program to zero the spindle to baseplate gap measurement, the stack of 40 mm diameter solid sheet discs are then to the rheometer spindle by gently flattening the bottom side of the sheet stack against the flat spindle surface. No adhesives are added as the sheet is sufficiently adhesive to remain adhered to the spindle surface.

[0228] Once the mass of sheet discs is adhered to the spindle, 0.62 g of deionized water is dispensed onto the center of the baseplate using an adjustable air-displacement pipette. For all tests the same mass of 0.62 g water was always added. The modified oscillation test program was then initiated.

[0229] A minimum of 3 experimental repeats are carried out for each sample type. The experimental time and experimental shear modulus (elastic component) data is then exported to Microsoft excel and the following parameters calculated: The peak shear modulus is the maximum observed value of the shear modulus measured in the experiment. The final shear modulus is the average value of the last 60 data points measured over the 10 minutes experimental time (for all experiments the relative standard deviation of this average value was less than 1.0%. If this is not the case the average value should not be considered as a correct estimate of the final shear modulus). The total area is the area under the shear modulus versus time curve and is calculated using the well known trapezoidal rule. Higher values for any of these three parameters indicate relatively slower dissolution of the sample.

Test 11: Visual Scoring Test for Leakage of the Coating Composition

[0230] The visual scores for leakage of coating composition applied on dissolvable sheets is measured as follows:

[0231] Once the samples are obtained (i.e., after cut-seal), a visual scoring test was performed to assess the degree of dye leakage on the top and bottom surface of the cut-sealed

multilayer samples. Each sample was firstly stored in an individual polyethylene zip-lock bag for three days. All samples were kept separate to avoid any compression force on the samples.

[0232] After three days the samples were removed from the polyethylene bags for measurement. Hollow, circular metal discs with radii ranging from 0.5 cm to 2.5 cm were used to quantify the leakage on the bottom and top surfaces of each sample by assigning a score from 0 to 5, as follows:

[0233] Score value=0—there are no obvious indications of leakage

[0234] Score value=1—the leakage area can be enclosed by a circle disk with internal radius of 0.5 cm

[0235] Score value=2—the leakage area can be enclosed by a circle disk with internal radius of 1.0 cm

[0236] Score value=3—the leakage area can be enclosed by a circle disk with internal radius of 1.5 cm

[0237] Score value=4—the leakage area can be enclosed by a circle disk with internal radius of 2.0 cm

[0238] Score value=5—the leakage area can be enclosed by a circle disk with internal radius of 2.5 cm

[0239] If the staining has occurred in multiple separate locations, the above method will be applied to each individual stained location. The scores for each individual location are summed and used as the final overall score.

Test 12: Viscosity

[0240] The viscosity values of liquid juice are measured using a Malvern Kinexus Lab+ rheometer with cone and plate geometry (CPI/50 SR3468 SS), a gap width of 0.054 mm, a temperature of 20° C. and a shear rate of 1.0 reciprocal seconds for a period of 360 seconds.

EXAMPLES

Example 1: Different OCF Structures in Solid Sheet Made by Different Heating/Dry Arrangements

[0241] Wet pre-mixtures with the following surfactant/polymer compositions as described in Table 1 and Table 2 below are prepared, for laundry care and hair care sheets, respectively.

TABLE 1

(LAUNDRY CARE FORMULATION)		
Materials:	(Wet) w/w %	(Dry) w/w %
Polyvinyl alcohol (with a degree of polymerization of about 1700)	7.58	21
Glycerin	1.08	3
Linear Alkylbenzene Sulfonate	19.12	53
Sodium Laureth-3 Sulfate	3.61	10
C12-C14 Ethoxylated alcohol	3.61	10
Water	Balance	Balance

[0242] Viscosity of the wet pre-mixture composition as described in Table 1 is about 14309.8 cps. After aeration, the average density of such aerated wet pre-mixture is about 0.25 g/cm³.

TABLE 2

(HAIR CARE FORMULATION-SHAMPOO)		
Materials:	(Wet) w/w %	(Dry) w/w %
Polyvinyl alcohol (with a degree of polymerization of about 1700)	6.85	23.69
Glycerin	2.75	9.51
Sodium Lauryl Sulfate	9.52	32.89
Sodium Laureth-3 Sulfate	3.01	10.42
Sodium Lauroamphoacetate	5	17.28
Citric acid (anhydrous)	0.93	3.21
Water	Balance	Balance

[0243] Viscosity of the wet pre-mixture composition as described in Table 2 is about 19254.6 cps. After aeration, the average density of such aerated wet pre-mixture is about 0.225 g/cm³.

[0244] Flexible, porous, dissolvable solid Sheets A and B are prepared from the above wet pre-mixtures as described in Tables 1 and 2 using a continuous aerator (Aeros) and a rotary drum drier, with the following settings and conditions as described in Table 3 below:

TABLE 3

(DRUM DRYING)	
Parameters	Value
Wet pre-mixture temperature before and during aeration	80° C.
Aeros feed pump speed setting	600
Aeros mixing head speed setting	500
Aeros air flow rate setting	100
Wet pre-mixture temperature before drying	60° C.
Rotary drum drier surface temperature	130° C.
Rotary drum drier rotational speed	0.160 rpm
Drying time	4.52 min

[0245] A flexible, porous, dissolvable solid Sheet C is also prepared from the above wet pre-mixture as described in Table 2 using a continuous aerator (Oakes) and a mold placed on a hot plate (which provides bottom conduction-based heating), with the following settings and conditions as described in Table 4 below:

TABLE 4

(HOT PLATE DRYING)	
Parameters	Value
Wet pre-mixture temperature before and during aeration	80° C.
Oakes air flow meter setting	19.2 L/hour
Oakes pump meter speed setting	20 rpm
Oakes mixing head speed	1500 rpm
Mold depth	1.0 mm
Hot plate surface temperature	130° C.
Drying time	12.5 min

[0246] Further, flexible, porous, dissolvable solid Sheets I and II are prepared from the above wet pre-mixtures described in Tables 1 and 2 using a continuous aerator

(Oakes) and a mold placed on an impingement oven, with the following settings and conditions as described in Table 5 below:

TABLE 5

(IMPINGEMENT OVEN DRYING)	
Parameters	Value
Wet pre-mixture temperature before and during aeration	80° C.
Oakes air flow meter setting	19.2 L/hour
Oakes pump meter speed setting	20 rpm
Oakes mixing head speed	1500 rpm
Mold depth	1.0 mm
Impingement oven temperature	130° C.
Drying time	6 min

[0247] Tables 6-9 as follows summarize various physical parameters and pore structures measured for the solid Sheets A-C and solid Sheets I-II made from the above-described wet pre-mixtures and drying processes.

TABLE 6

(PHYSICAL PARAMETERS)						
Sheet	Formu- lation	Drying Process	Average Basis Weight g/m ²	Average Density g/cm ³	Average Thick- ness mm	Specific Surface Area m ² /g
A	Laundry Care	Rotary Drum	147.5	0.118	1.265	0.115
B	Hair Care	Rotary Drum	138.4	0.111	1.254	0.107
C	Hair Care	Hot Plate	216.3	0.111	1.968	—
I	Laundry Care	Impingement Oven	116.83	0.118	1.002	—
II	Hair Care	Impingement Oven	212.9	0.111	1.929	—

TABLE 7

(OVERALL PORE STRUCTURES)					
Sheet Samples	Formulation	Drying Process	Percent Open Cell Content %	Overall Average Pore Size µm	Average Cell Wall Thickness µm
A	Laundry Care	Rotary Drum	90.75	467.1	54.3
B	Hair Care	Rotary Drum	93.54	466.9	42.8
C	Hair Care	Hot Plate	—	287.4	19.7
I	Laundry Care	Impingement Oven	—	197.6	15.2
II	Hair Care	Impingement Oven	—	325.1	18.7

TABLE 8

(SURFACE AND REGIONAL PORE STRUCTURES)						
Sheet	Formu-	Drying	Surface Average Pore Diameter (μm)	Average Pore Size (μm)		
Samples	lation	Process	Top	Top	Middle	Bottom
A	Laundry Care	Rotary Drum	201.5	458.3	479.1	463.9
B	Hair Care	Rotary Drum	138.2	412.4	519.0	469.1
C	Hair Care	Hot Plate	120.8	259.7	292.0	309.9
I	Laundry Care	Impingement Oven	53.3	139.9	213.1	238.7
II	Hair Care	Impingement Oven	60.0	190.7	362.6	419.6

TABLE 9

(VARIATIONS BETWEEN REGIONAL PORE STRUCTURES)						
Sheet	Formu-	Drying	Cross- Region	Btw-Region Ratios of Average Pore Sizes		
Sam- ples	lation	Process	Relative STD (%)	Bottom- to-Top	Bottom- to-Middle	Middle- to-Top
A	Laundry Care	Rotary Drum	2.31%	1.012	0.968	1.046
B	Hair Care	Rotary Drum	11.43%	1.137	0.904	1.259
C	Hair Care	Hot Plate	8.84%	1.193	1.061	1.124
I	Laundry Care	Impingement Oven	25.99%	1.706	1.120	1.523
II	Hair Care	Impingement Oven	36.74%	2.200	1.157	1.901

[0248] The above data demonstrates that the solid sheets of the present invention as being predominantly open-celled and that the solid sheets made by the rotary drum-drying process have Top Surface Average Pore Diameters of greater than 100 μm , while the solid sheets made by the impingement oven process do not. Specifically, FIG. 6A shows a Scanning Electron Microscopic (SEM) image of the top surface of the Sheet A, while FIG. 6B shows a SEM image of the top surface of the solid Sheet I. FIG. 7A shows a SEM image of the top surface of the solid Sheet C, while FIG. 7B shows a SEM image of the top surface of the solid Sheet II.

[0249] Further, the above data demonstrates that the solid sheets made by the rotary drum-drying process have significantly less regional variations in their Average Pore Sizes than the solid sheets made by the impingement oven process, especially with significantly smaller ratios of the bottom Average Pore Size over the top Average Pore Size.

Example 2: Improved Dissolution Profile of Solid Articles with Juice Compared to Solid Articles without Juice

1) Preparation of Solid Articles with Juice and Solid Articles without Juice

[0250] Dissolvable solid articles containing the coating composition (hereinafter referred to as Solid Articles with Juice) and dissolvable solid articles without the coating composition (hereinafter referred to as Solid Articles without Juice) are prepared as follows.

[0251] Firstly, large solid sheets (with minimum area 1.0×1.0 m) are prepared according to the method in the Section III: PROCESS OF MAKING SOLID SHEETS.

[0252] Specifically, a wet pre-mixture containing the ingredients of solid sheets and additional water is first prepared, to result in a total solids content of about 35% by weight (i.e., the total water content in the slurry is about 65% by weight). The method of slurry preparation is as follows:

[0253] 1. Water and glycerin are firstly added together into a glass beaker and stirred at 200 rpm using an overhead stirrer.

[0254] 2. While continuing to stir, the polyvinyl alcohol is then slowly added into the beaker containing water and glycerin, ensuring that no foaming of the solution or clumping of the polyvinyl alcohol occurred.

[0255] 3. The beaker is then placed in a water bath and heated to 80° C. while continuing stirring. The beaker is covered with clingfilm or tinfoil in order to mitigate water evaporation and left to continue mixing for at least 1.0 hour.

[0256] 4. The remaining components are weighed and added together in a separate glass beaker. The balance of water required to achieve 65% total water content in the slurry is also added to this beaker.

[0257] 5. This beaker is placed in a water bath at 80° C., and its contents are stirred using an overhead stirrer at 500 rpm for at least 30 minutes.

[0258] 6. Once the predefined mixing time is reached in both beakers, the contents of both are added together into a single glass beaker, followed by continued stirring at 500 rpm and the temperature is maintained at 80° C. for at least another 30 minutes.

[0259] The wet pre-mixture so formed has a viscosity of about 19254.6 cps. It is then aerated as follows:

[0260] 1. An Aeros A20 continuous aerator, consisting of a jacketed hopper (model JCABT10) and A20 mixing head, is preheated to 80° C. using a water bath and pump.

[0261] 2. The slurry prepared previously is then added to the hopper. The aerator unit is then switched on and the mixing head speed, feed pump speed, and air flow rates were set to 600, 500 and 100 respectively.

[0262] 3. The aerated slurry is collected from the aerator outlet and its density measured by filling a density cup of known volume and weighing the mass of the aerated slurry. At the aerator settings described above, an aerated slurry density of about 0.225 g/cm³ is achieved.

[0263] Flexible and porous solid sheets of about 0.8-1.5 mm in thickness are produced using a rotary drum dryer process, as follows:

[0264] 1. The rotary drum dryer (having a drum diameter of about 1.5 m) is pre-heated to about 130° C.

[0265] 2. The aerated slurry collected from the Aeros A20 outlet is added to the feeding trough of the drum dryer.

[0266] 3. Once added, the rotation of the drum dryer starts and is set at a rotating speed so that the slurry residence time on the heated drum is about 15 minutes.

[0267] 4. Once dried, the flexible and porous sheets so formed are peeled from the drum surface and placed in a plastic bag.

[0268] Then, the solid sheets are stored under ambient relative humidity of 50±2% and temperature of 23±1° C. for 24 hours (i.e., a conditioning step). The average thickness of

all the Sheet 1 is 1.2107 mm with a standard deviation of 0.0464. Following the initial conditioning step described above, 4 cm diameter discs are firstly cut from the large solid sheet using a 4 cm hollow hole punch. Then, the coating composition is added according to Coating Method A as described below if the coating composition is required.

[0269] In Coating Method A, a pipette is employed to dispense droplets of the coating composition onto a single location on the surface of the solid sheets. This location is always the centermost point of the total foam mass. FIG. 8A illustrates an exemplary solid article obtained by using the Coating Method A. For example, if there is a single solid sheet required in the experiment, the droplets are dispensed onto the centermost point on the bottom surface of that solid sheet. If multiple solid sheets are required in the experiment, half of the sheets are firstly stacked in head-to-toe configuration, the coating composition is then dispensed onto the centermost point of the top sheet, and the remaining sheets then stacked on top. For a single sheet or multiple stacked sheets, the sheets are always orientated such that the coating composition is dispensed onto the bottom side of the sheet. With the solid sheet placed on a mass balance and the mass tared to zero, the droplets are continuously added until the required mass of coating composition is achieved.

[0270] Then, the samples are stored for another 24 hours after the addition of the coating composition at the same humidity and temperature conditions ($50\pm 2\%$ and $23\pm 1^\circ\text{C}$.).

[0271] The sheets and the coating composition respectively have the formulation shown in the following tables:

TABLE 10

(SHEET FORMULATION)	
Materials (Dry) w/w %	Sheet 1
Polyvinyl alcohol (with a degree of polymerization of about 1700)	18.00
Polyvinyl alcohol (with a degree of polymerization of about 500)	6.00
Glycerin	3.51
Linear Alkylbenzene Sulfonate	40.00
Sodium Laureth-3 Sulfate	4.60
C12-C14 Ethoxylated alcohol	16.00
Ethoxylated Polyethyleneimine	1.50
Palm kernel fatty acid soap powder	2.07
Sodium Aluminosilicate (crystalline)/Zeolite	0.95
Denatonium Benzoate	0.04
Water	6.00
Miscellaneous	Q.S.

TABLE 11

(JUICE FORMULATION)		
Materials (w/w %)	Juice 1 (w/o Solvent)	Juice 2 (with Solvent)
C12-C14 Ethoxylated alcohol	62.00	55.80
Perfume	38.00	34.20
Dipropylene Glycol	0.00	10.00
Viscosity ($\text{Pa} \cdot \text{s}$, at about 20°C . and 1 s^{-1})	0.0033	0.0039

[0272] Three types of samples: Articles 1 to 3 (three replicates per type) are prepared as shown in the following table.

TABLE 12

	Article 1 Sheet 1 only	Article 2 Sheet 1 + Juice 1 (w/o Solvent)	Article 3 Sheet 1 + Juice 2 (with Solvent)
Total mass, gram	0.691	0.694	0.735
Sheet mass, gram	0.691	0.444	0.450
Juice mass, gram	—	0.250	0.286
Total mass of surfactants, gram	0.418	0.424	0.431
Total mass of PVA, gram	0.166	0.107	0.108

[0273] Particularly, Article 1 is formed by stacking three layers of Sheet 1 without applying any coating composition; Article 2 is formed by adding Juice 1 on one layer of Sheet 1 using Coating Method A mentioned hereinabove and then adding another layer of Sheet 1 on top to form a 2-layer stack; and Article 3 is formed by adding Juice 2 on one layer of Sheet 1 using Coating Method A and then adding another layer of Sheet 1 on top to form a 2-layer stack. Lastly, for all samples, the amount of coating compositions added is calculated so that the total mass of surfactant in the sample (from both the sheet and the juice) was equal to approximately 0.42 grams.

[0274] 2) Measurement of Foam Gelling Gelling occurs when the solid articles according to the present disclosure are contacted with water due to the dissolution of water-soluble polymer (e.g., PVA) and surfactants in the solid articles. The presence of gelling might prevent the water to penetrate into the solid articles through the OCF structure, resulting in the reduced dissolution rate. Furthermore, once a hard gel is formed it would be very slow for the solid article to further dissolve, likely resulting in residues on clothes if the solid article is used for laundry. As such, if gelling degree is reduced, the dissolution profile is improved.

[0275] Gelling of Solid Article without Juice (Article 1) and Solid Articles with Juice (Articles 2 and 3) is determined according to Test 10. The results are shown as below.

TABLE 13

	Article 1 Sheet only	Article 2 Sheet + Juice (w/o Solvent)	Article 3 Sheet + Juice (with Solvent)
Shear modulus G' peak value, Pa	9925	2729	954
Shear modulus G' final value, Pa	3411	1563	1205
Total Area, Pa	2143562	1341347	652528

[0276] The above data shows that significantly higher values for the three measured parameters (shear modulus G' peak, G' final and total area) are observed for the solid sheet only sample, indicating worse dissolution. The results of the gelling test are also shown in FIG. 9. It is completely surprising that Solid Articles with Juice (e.g., Articles 2 and 3) show improved dissolution profile compared to Solid Article without Juice (e.g., Article 1), because it was believed prior to the filing of the present disclosure that loading of a coating composition on the solid article according to the present disclosure might compromise the dissolution by blocking the OCF structure.

[0277] Furthermore, a significant reduction is also observed for the shear modulus peak value of the sample

containing the coating composition with added solvent (Article 3), compared to the sample without a solvent in the juice (Article 2), indicating that including a solvent in the coating composition may bring about an even more improved dissolution profile (e.g., even less gelling).

Example 3: Juice Loading Capacity of the Solid Article without Leakage

1) Preparation of Two Series of Multilayer Sheets Containing Different Amounts of the Coating Composition

[0278] Two series of multilayer sheets containing different amounts of the coating composition applied by different coating methods are prepared, in which Series 1 is prepared from Sheet 1 and Juice 1 as mentioned in Example 2, and Series 2 is prepared from Sheet 1 and Juice 3 (containing silicon dioxide as a rheology modifier) as shown in the following table. The mass of added juices across these samples range from approximately 2 to 12 g. The average estimated sheet density of these multilayer samples is 0.169 g/cm³.

TABLE 14

(JUICE FORMULATION)	
Materials (w/w %)	Juice 3 (with rheology modifier)
C12-C14 Ethoxylated alcohol	58.00
Perfume	30.50
Silicon dioxide	11.50
Viscosity (Pa · s, at about 20° C. and 1 s ⁻¹)	4.5257

[0279] The preparation of such two series of multilayer sheets is the same with that in Example 2, except that after the initial conditioning step, the large solid sheet is firstly cut to 10×10 cm sheets using a paper guillotine, followed by the addition of the coating composition according to Coating Method A as mentioned in Example 2 or Coating Method B as below.

[0280] In Coating Method B, a plastic roller (with roller width of 10 cm and diameter 2 cm) is employed to spread the coating composition across a 10×10 cm solid sheet. The roller is firstly rolled across a walled container having a flat surface larger than 10×10 cm and containing a pool of the liquid. Excess liquid is then removed by gently shaking the roller. The roller is then rolled a minimum of 10 times throughout the 10×10 cm sheet, where the initial point of contact between the roller and sheet and the direction of rolling is randomized each time in order to help prevent inhomogeneous coating. The coating composition is always rolled across the bottom side of the sheet. With the 10×10 cm solid sheet placed on a mass balance and the mass tared to zero, this procedure is repeated on the bottom side of sheets until the required mass of coating composition is spread onto the sheet surface. However, no juice is applied onto the top sheet and bottom sheet of the stack, in order to act as a buffer against leakage, as shown in FIG. 8B.

[0281] All multilayer samples as prepared by the Coating Method A in this example are comprised of eighteen stacked layers of 10×10 cm solid sheets, and all multilayer samples as prepared by the Coating Method B in this example are comprised of thirteen stacked layers. For this example, the coating composition comprises 0.1 wt % of a dye (Liquitint

Violet 129), and the content of perfume is correspondingly reduced by 0.1 wt %. Furthermore, once the coating composition is added, the multilayer stack is cut-sealed by using a Chhong 1 tonne CH217 hydraulic press (S/N HP170726TJ01) with a cut angle ranging from about 200 to about 50°. The cutting blade used for the cut-seal is an irregular closed shape with an internal area of 3182 mm². The mass of each sheet after cutting was weighed at 0.57 g with a standard deviation of 0.019 g. For the cut-seal samples, the final mass of the added coating composition was estimated by the following formula: Coating composition mass=Total mass of cut-seal sample-18*0.57. This equation was used to account for some samples with high coating composition loading, wherein excessive leakage occurred and some of the added coating composition mass leaked onto the cut-seal equipment.

2) Measurement of Leakage Score

[0282] Then, the leakage scores are determined according to Test 11. The results are shown in the following table.

TABLE 15a

Series 1: Sheet 1 + Juice 1 Average Leakage Scores		
Juice mass, gram	Coating Method A	Coating Method B
Lower than 2.0	0.0	0.0
Between 2.0 and 4.0	0.5	0.5
Between 4.0 and 6.0	5.7	5.3
Between 6.0 and 8.0	7.3	6.0
Greater than 8.0	N/A	9.0

TABLE 15b

Series 2: Sheet 1 + Juice 3 (with rheology modifier) Average Leaking Scores		
Juice mass, gram	Coating Method A	Coating Method B
Between 2.0 and 4.0	0.0	0.0
Between 4.0 and 6.0	1.8	0.0
Between 6.0 and 8.0	3.3	1.3
Between 8.0 and 10.0	4.0	1.3
Greater than 10.0	N/A	3.5

[0283] It indicates that a significant mass of the coating composition may be applied without significant leakage. Particularly, regarding Series 1, the results show that no significant leakage (i.e., the score is less than 1) when the added juice is less than 4.0 g. And, regarding Series 2, the results clearly show that for Coating Method A no leakage (i.e., the score is 0) is observed below 4.0 g of added liquid juice, and for Coating Method B no leakage is observed below 6.0 g of added liquid juice. Additionally, the results show that the leakage score is consistently lower for Coating Method B across the entire range of added juice mass.

[0284] Furthermore, the results indicate that a preferred coating composition (e.g., Juice 3) brings about an even more improved anti-leakage performance.

Example 4: Effect of Foam Structure on Juice Leakage

1) Preparation of High-Density and Low-Density Multilayer Sheets Containing the Coating Composition

[0285] Sheets having the same composition (as shown hereinbelow in Table 16) but different densities were prepared from the same wet pre-mix by changing the target density of the aerated wet premix on the continuous aerator to 0.3 g/cm³ and 0.4 g/cm³ for the lower and higher density sheets, respectively. The high-density sheets have an average estimated foam density of 0.177 g/cm³, and the low-density sheets have an average density of 0.135 g/cm³.

TABLE 16

(SHEET FORMULATION)	
Materials (Dry) w/w %	
Polyvinyl alcohol (with a degree of polymerization of about 1700)	18.00
Glycerin	9.00
Linear Alkylbenzene Sulfonate	56.00
Sodium Laureth-3 Sulfate	6.00
Ethoxylated Polyethyleneimine	2.00
Palm kernel fatty acid soap powder	2.00
Water	6.00
Miscellaneous	Q.S.

[0286] Multilayer sheets containing the coating composition are prepared from the high-density sheets or the low-density sheets as mentioned hereinabove with added Juice 3 as mentioned in Example 3. Particularly, 3 samples of multilayer sheets are prepared, each containing thirteen layers of the high-density sheets with Juice 3 added therein according to Coating Method B as mentioned in Example 3, which are referred to herein as Articles 4. Similarly, 3 samples of multilayer sheets are prepared, each containing eighteen layers of the low-density sheets with Juice 3 added therein according to Coating Method B as mentioned in Example 3, which are referred to herein as Articles 5. The total thickness for both Articles 4 and Articles 5 is maintained as 20 mm by modifying the average thickness of each sheet.

2) Measurement of Leakage Score

[0287] Then, the average leakage scores of Articles 4 and Articles 5 are determined according to Test 11. The results are shown in the following table.

TABLE 17

	Average sheet mass	Average juice mass	Average leakage score
Articles 4	11.25	5.45	4.67
High density (0.177 g/cm ³)			
Articles 5	8.60	5.45	0.67
Low density (0.135 g/cm ³)			

[0288] The results indicate that the Articles 5 made with low-density sheets shows no significant leakage (score=0.67) while the Articles 4 made with high-density sheets experiences leakage of juice (score=4.67), indicating that

the density of the sheets is important for preventing leakage and particularly, lower density sheets can hold more juice before leakage occurs.

Example 5: No Significant Blocking of OCF Structure by Juice

1) Preparation of Single Layer Sheets Containing the Coating Composition

[0289] A single-layer Article 6 is prepared from Sheet 1 only as mentioned in Example 2, without any added juice. Another single-layer Article 7 is prepared from Sheet 1 and Juice 3 as mentioned in Examples 2 and 3. Particularly, Juice 3 is added to Sheet 1 according to Coating Method B except that, instead of rolling the liquid juice onto the bottom side of Sheet 1, it is rolled onto the top side. Because the top side is even more porous than the bottom side, the top side would be more suitable to show if the juice results in significant pore blockage. An average of 2.8 grams of liquid juice is added with standard deviation of 0.3 grams.

2) SEM Test

[0290] SEM test is conducted according to Test 1 to visualize the top surfaces of Articles 6 and 7. FIG. 10A shows the top surface of Article 6. FIG. 10B shows the top surface of Article 7. It is evident that even after applying a high juice loading (around 2.8 g), the OCF structure of Article 7 is not significantly compromised (i.e., not blocked by the juice).

Example 6: Exemplary Solid Articles with Juice

[0291] The following are examples of Solid Articles with Juice. The Sheets a to e (see Table 18) are prepared similarly as Sheet 1 in Example 2. Then, Solid Articles with Juice are formed by applying Juices a to e (see Table 19) to Sheets a to e according to Coating Method A or B and then stacking the respective sheets to form multilayer structures with 10-20 layers each. Such articles may be respectively used for laundry and personal cleansing care or hair care (PCC/Hair).

TABLE 18

(SHEET FORMULATION)					
Materials (Dry) w/w %	Sheet a	Sheet b	Sheet c	Sheet d	Sheet e
		Laundry		PCC/Hair	
Polyvinyl alcohol (with a degree of polymerization of about 2400)	0.00	0.00	12.00	0.00	0.00
Polyvinyl alcohol (with a degree of polymerization of about 1700)	18.00	18.00	0.00	23.72	24.00
Polyvinyl alcohol (with a degree of polymerization of about 500)	6.00	0.00	0.00	0.00	0.00
Glycerin	3.51	9.00	9.00	9.04	9.15
Linear Alkylbenzene Sulfonate	40.00	56.00	56.00	0.00	0.00
Sodium Lauryl Sulfate	0.00	0.00	6.00	36.53	36.97
C12-C14 Ethoxylated alcohol	16.00	0.00	0.00	0.00	0.00
Sodium Laureth-3 Sulfate	4.60	6.00	6.00	9.91	10.03
Sodium Lauroamphoacetate	0.00	0.00	0.00	11.16	11.30

TABLE 18-continued

(SHEET FORMULATION)					
Materials (Dry) w/w %	Sheet a	Sheet b	Sheet c	Sheet d	Sheet e
		Laundry		PCC/Hair	
Ethoxylated Polyethyleneimine	1.50	2.00	2.00	0.00	0.00
Palm kernel fatty acid soap powder	2.07	2.00	2.00	0.00	0.00
Sodium Aluminosilicate (crystalline)/Zeolite	0.95	0.00	0.00	0.00	0.00
Denatonium Benzoate	0.04	0.00	0.00	0.00	0.01
Sodium Benzoate	0.00	0.00	0.00	0.45	0.45
Citric Acid	0.00	0.00	0.00	2.07	2.09
Perfume	0.00	0.00	0.00	1.12	0.00
Water	6.00	6.00	6.00	6.00	6.00
Miscellaneous	1.33	1.00	1.00	0.00	0.00

TABLE 19

(JUICE FORMULATION)					
Materials (w/w %)	Juice a	Juice b	Juice c	Juice d	Juice e
		Laundry		PCC/Hair	
C12-C14 Ethoxylated alcohol	58.00	55.80	62.50	20.00	0.00
Sodium Laureth-3 Sulfate	0.00	0.00	12.50	0.00	0.00
Perfume	30.50	34.20	18.75	45.00	23.45
Silicon dioxide	11.50	0.00	0.00	7.50	0.00
Dipropylene Glycol	0.00	10.00	0.00	0.00	4.65
1,2-propane diol	0.00	0.00	0.00	0.00	10.56
Sodium Laureth-1 Sulfate	0.00	0.00	0.00	0.00	53.65
Ethoxylated Polyethyleneimine	0.00	0.00	6.25	0.00	0.00
Citric Acid	0.00	0.00	0.00	0.00	1.74
Glycerin	0.00	0.00	0.00	0.00	5.95
Perfume Oil Capsules	0.00	0.00	0.00	7.50	0.00
Water	0.00	0.00	0.00	20.00	0.00

[0292] The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as “40 mm” is intended to mean “about 40 mm.”

[0293] Every document cited herein, including any cross reference or related patent or application and any patent application or patent to which this application claims priority or benefit thereof, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

[0294] While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to

cover in the appended claims all such changes and modifications that are within the scope of this invention.

What is claimed is:

1. A process for preparing a dissolvable solid article comprising the steps of:

- 1) providing two or more flexible, porous, dissolvable sheets and a coating composition, wherein each of said two or more sheets comprises a water-soluble polymer and a first surfactant and is characterized by a Percent Open Cell Content of from about 80% to about 100% and an Overall Average Pore Size of from about 100 μm to about 2000 μm , and wherein said coating composition comprises a second surfactant;
- 2) applying the coating composition on at least one surface of at least one sheet from said two or more sheets; and
- 3) arranging the two or more sheets into a stack to form the dissolvable solid article, wherein the stack has contact surfaces where each of the two or more sheets touch and outer surfaces wherein at least a portion of the two or more sheets do not touch, so that the coating composition is not on any of the outer surfaces of the stack.

2. The process of claim 1, wherein at least one of said two or more sheets comprises from about 5% to about 50%, of said water-soluble polymer by total weight of said sheet;

wherein said water-soluble polymer has a weight average molecular weight of from about 50,000 to about 400,000 Daltons; and

wherein said water-soluble polymer is a polyvinyl alcohol characterized by a degree of hydrolysis ranging from about 40% to about 100%.

3. The process of claim 1, wherein at least one of said two or more sheets comprises from about 50% to about 70%, of said first surfactant by total weight of said sheet; wherein said first surfactant is selected from the group consisting of a $\text{C}_6\text{-C}_{20}$ linear alkylbenzene sulfonate, a $\text{C}_6\text{-C}_{20}$ linear or branched alkylalkoxy sulfate having a weight average degree of alkoxylation ranging from about 0.5 to about 10, a $\text{C}_6\text{-C}_{20}$ linear or branched alkyl sulfate, a $\text{C}_6\text{-C}_{20}$ linear or branched alkylalkoxylated alcohol having a weight average degree of alkoxylation ranging from about 5 to about 15, and any combination thereof.

4. The process of claim 1, wherein said coating composition is a liquid having a viscosity from about 3 cps to about 5,000 cps, as measured at 20° C. and 1 s^{-1} .

5. The process of claim 1, wherein said second surfactant comprises a non-ionic surfactant comprising a $\text{C}_6\text{-C}_{20}$ linear or branched alkylalkoxylated alcohols having a weight average degree of alkoxylation ranging from about 5 to about 15.

6. The process of claim 1, wherein said coating composition further comprises a solvent selected from the group consisting of glycerol, propylene glycol, 1,3-propanediol, diethylene glycol, dipropylene glycol, ethanolamine, ethanol, water, and any combination thereof.

7. The process of claim 1, wherein said coating composition further comprises a rheology modifier selected from the group consisting of: cellulose, a guar, polyethylene oxide, polypropylene oxide, polyethylene oxide/polypropylene oxide copolymers; polyvinylpyrrolidone, crosslinked polyvinylpyrrolidone, polyvinylalcohol, polyethyleneimine; sodium carbonate, sodium sulphate, silicon dioxide, water-swelling clays, gums, and any combination thereof.

8. The process of claim 1 wherein said coating composition further comprises a perfume, wherein the ratio by weight of said second surfactant to said perfume in said coating composition is from about 1:50 to about 50:1.

9. The process of claim 1, wherein said coating composition comprises:

- 1) from about 10% to about 80%, of said second surfactant by total weight of said coating composition;
- 2) from about 1% to about 50%, of said solvent by total weight of said coating composition;
- 3) from about 1% to about 50%, of said rheology modifier by total weight of said coating composition; and
- 4) from about 10% to about 80%, of said perfume by total weight of said coating composition.

10. The process of claim 1, wherein said coating composition further comprises an additional ingredient selected from the group consisting of a silicone, a softening agent, a bleaching agent, an enzyme, an anti-bac agent, an anti-oxidant, a brightener, a hueing dye, a polymer, a personal care active, and any combination thereof.

11. The process of claim 1, wherein said coating composition comprises less than 20%, of water by total weight of said coating composition.

12. The process of claim 1, wherein said coating composition is applied in an amount of about 10% to about 60%, by total weight of said dissolvable solid article.

13. The process of claim 1, wherein said coating composition is applied on one or more contacting surfaces of the sheets in said stack.

14. The process of claim 1, wherein said coating composition is applied on one or both contacting surfaces of any two adjacent sheets in said stack excluding the two outermost sheets.

15. The process of claim 1, wherein said coating composition is applied in a central region on each of an applied surface of the respective sheets, wherein the applied surface is a region that is spaced apart from the peripherals of the respective sheets by a distance that is at least 10%, of the maximum Dimension D.

16. A dissolvable solid article comprising two or more flexible, porous, dissolvable sheets, wherein each of said two or more sheets comprises a water-soluble polymer and

a first surfactant and is characterized by a Percent Open Cell Content of from about 80% to about 100% and an Overall Average Pore Size of from about 100 μm to about 2000 μm ; wherein the dissolvable solid article has an outer surface; and

wherein a coating composition comprising a second surfactant is present on at least one surface of at least one of said two or more sheets, provided that said coating composition is not on any of the outer surfaces of the dissolvable solid article.

17. The dissolvable solid article of claim 16, wherein at least one of said two or more sheets comprises from about 11% to about 25%, of said water-soluble polymer by total weight of said sheet;

wherein said water-soluble polymer has a weight average molecular weight from about 80,000 to about 150,000 Daltons; and

wherein said water-soluble polymer is a polyvinyl alcohol characterized by a degree of hydrolysis ranging from about 65% to about 92%.

18. The dissolvable solid article of claim 16, wherein at least one of said two or more sheets further comprises from about 0.00001% to about 1% of a bittering agent by total weight of said sheet.

19. The dissolvable solid article of claim 16, wherein said coating further comprises a perfume and the ratio by weight of said second surfactant to said perfume is from about 1:50 to about 50:1.

20. The dissolvable solid article of claim 16, wherein said coating composition comprises:

- 1) from about 10% to about 80%, of said second surfactant by total weight of said coating composition;
- 2) from about 1% to about 50%, of a solvent by total weight of said coating composition;
- 3) from about 1% to about 50%, of a rheology modifier by total weight of said coating composition; and
- 4) from about 10% to about 80%, of a perfume by total weight of said coating composition.

* * * * *